



Lecture:6



Major Theories of Learning

TATSIOL TUCOLIG2 OL TCSLUIUD

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• Major Learning Theories in Psychology

- During the first half of the twentieth century, the school of thought known as **behaviorism** rose to dominate psychology and sought to explain the learning process.
- The three major types of learning described by behavioral psychology are **classical conditioning**, **operant conditioning** and **observational learning**.

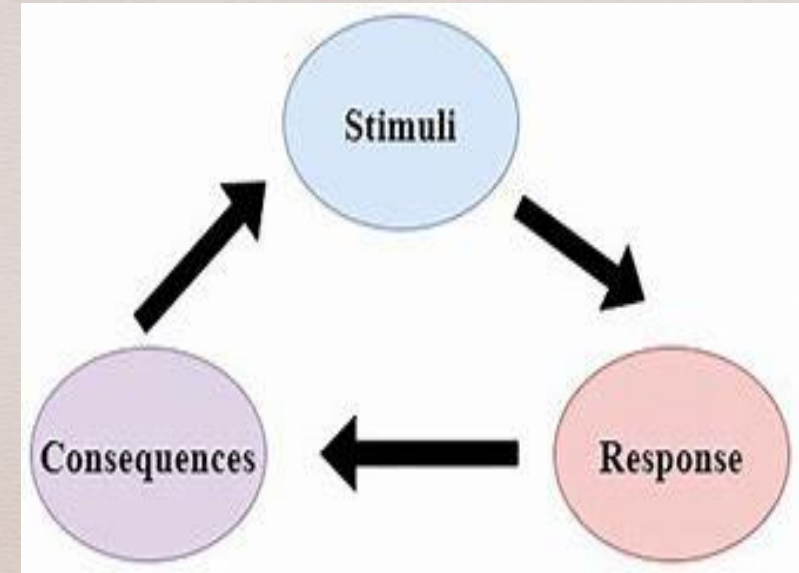


• What is Behaviorism?

- Behaviorism was the **school of thought in psychology** that sought **to measure only observable behaviors**.
- Founded by **John B. Watson** and outlined in his seminal 1913 paper *Psychology as the Behaviorist View It*,
- the behaviorist standpoint held that **psychology was an experimental** and objective **science** and that **internal mental processes should not be considered** because they could not be directly observed and measured.



- Behaviorism is primarily concerned with **observable and measurable** aspects of **human behavior**
- Behavior is **directed by stimuli**; changes in behavior that result from **stimulus-response associations**
- Only **behaviors worthy of study** are those that can be directly **observed**; does not explain abnormal behavior
- All **behaviors can also be unlearned**, and **replaced by new behaviors**



- J.B. Watson, B.F. Skinner, Ivan Pavlov are the pioneers of Behaviorism



Figure 1.1 Classical Conditioning (Ivan Pavlov: 1849-1936)



Figure 1.1. This illustration shows the steps of Classical Conditioning.

1. Food = salivation
2. Food + Stimulus = salivation (conditioned stimulus)
3. Bell alone produces salivation (conditioned response)

Classical Conditioning example

Before conditioning

**FOOD
(UCS)**

**SALIVATION
(UCR)**



BELL

NO RESPONSE



During conditioning

**BELL +
FOOD
(UCS)**

**SALIVATION
(UCR)**



Key terms

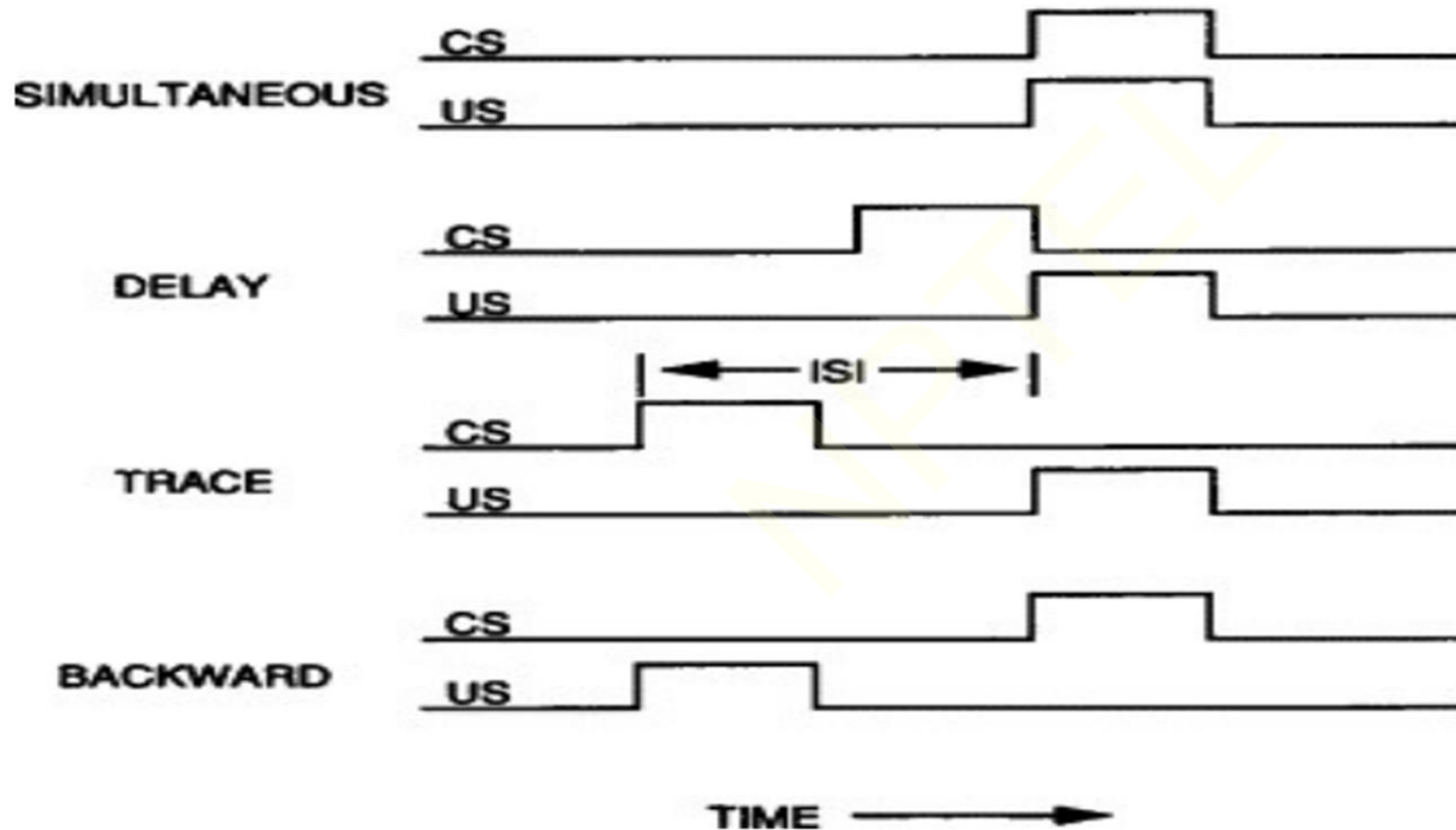
- Stimulus-physical event
- Unconditioned stimulus
- Unconditioned response
- Conditioned stimulus
- Conditioned response



conditioning



SIMPLE CONDITIONING PROCEDURES



Different procedures of classical conditioning

- **Delay conditioning:** A form of forward conditioning in which the onset of the unconditioned stimulus (UCS) begins while the conditioned stimulus (CS) is still present.
- **Trace conditioning:** A form of forward conditioning in which the onset of the conditioned stimulus (CS) precedes the onset of the unconditioned stimulus (UCS) and the presentation of the CS and UCS does not overlap.
- **Simultaneous conditioning:** A form of conditioning in which the conditioned stimulus (CS) and the unconditioned stimulus (UCS) begin and end at the same time.
- **Backward Conditioning:** A type of conditioning in which the presentation of the unconditioned stimulus (UCS) precedes the presentation of the conditioned stimulus (CS).



Principles of Classical Conditioning

- **Acquisition**
- **Extinction**
- **Reconditioning**
- **Spontaneous recovery**





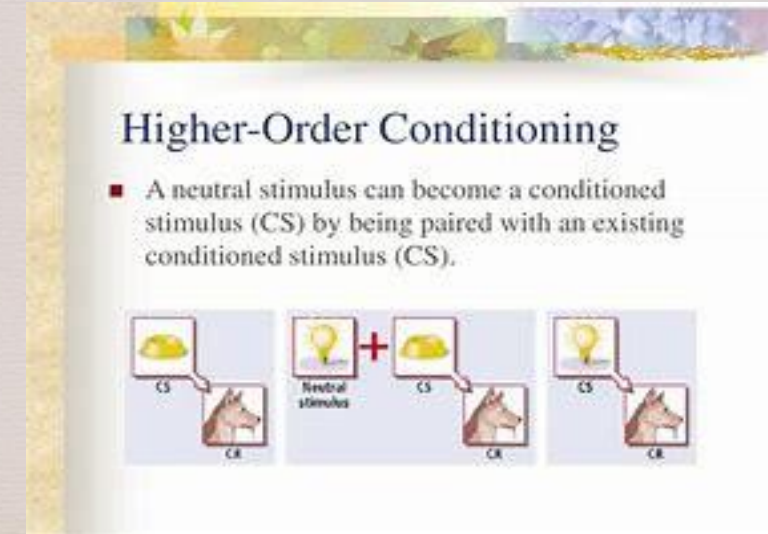
Principles of Classical Conditioning

- **Stimulus generalization:** The tendency of stimuli similar to a conditioned stimulus to evoke conditioned response.
- **Stimulus discrimination:** The process by which organisms learn to respond to certain stimuli but not to others



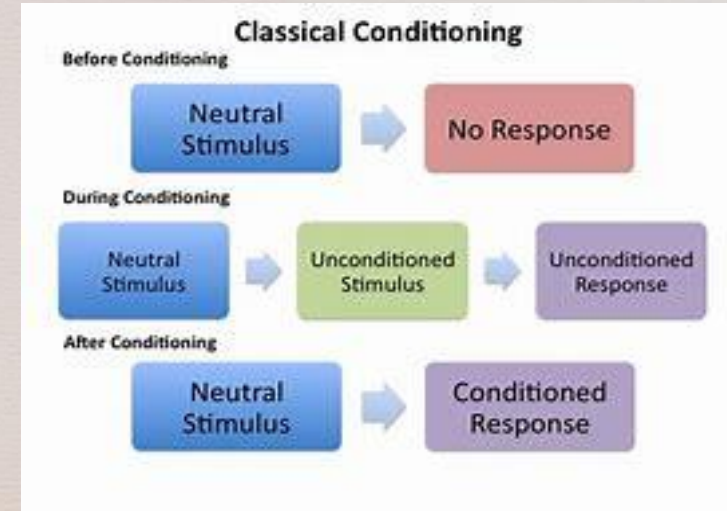
Higher Order Conditioning

- First-order – learning is obtained by associating a neutral stimulus with an unconditioned stimulus.
- Second-order – learning by pairing a neutral stimulus with a stimulus that has previously been converted into a conditioned stimulus through first-order conditioning.
- Higher-order – learning acquired by pairing a neutral stimulus with another stimulus previously conditioned. Higher-order conditioning is intrinsically weaker than its first-order counterpart. But the strength can be increased by using a robust unconditioned stimulus.



Classical conditioning: Day-to-day life examples

- A parent can turn “doing homework” into a game. Playing games (US) are fun (UR). After repeatedly paired together, doing homework (CS) and having fun (CR) create a positive association in the child.
- If a teacher is warm and nurturing (US), students feel connected and attached (UR). Students then associate going to school (CS) with connected feelings (CR) and learn to like going to school.
- If students are friendly and helpful to one another (US), students find it fun (UR) to go to school. Students associate going to school (CS) with fun (CR) and look forward to attending every day.



Expanding on Watson's basic stimulus-response model, Skinner developed operant conditioning based on the premise that satisfying responses are conditioned



Figure 1.2 Operant Conditioning (B. F. Skinner: 1904-1990)

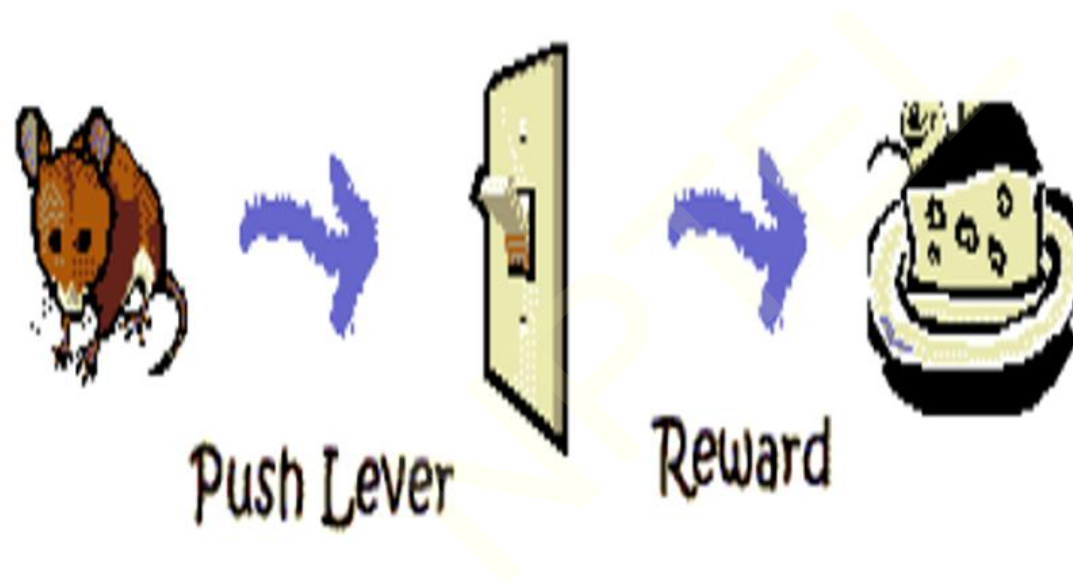
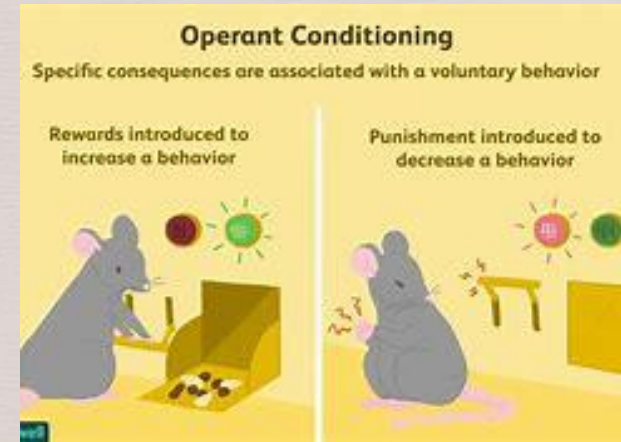
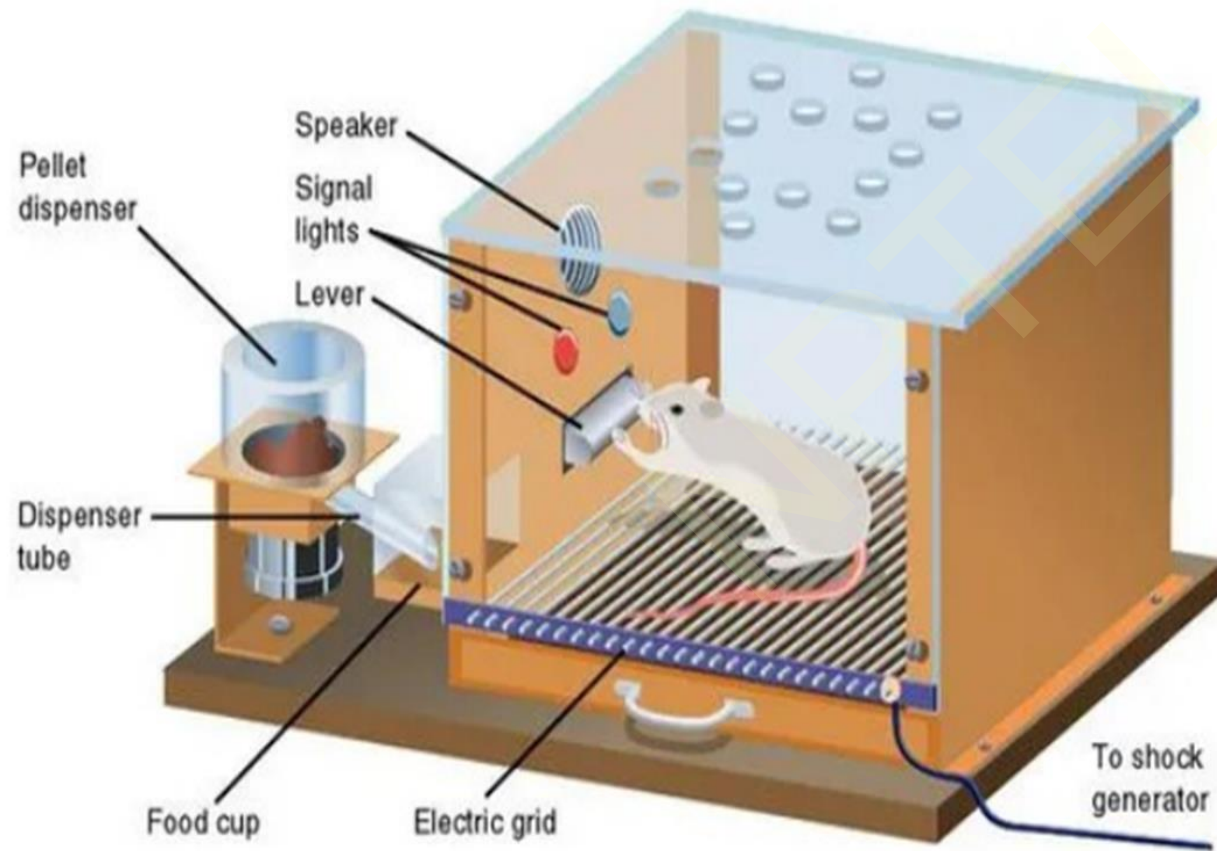
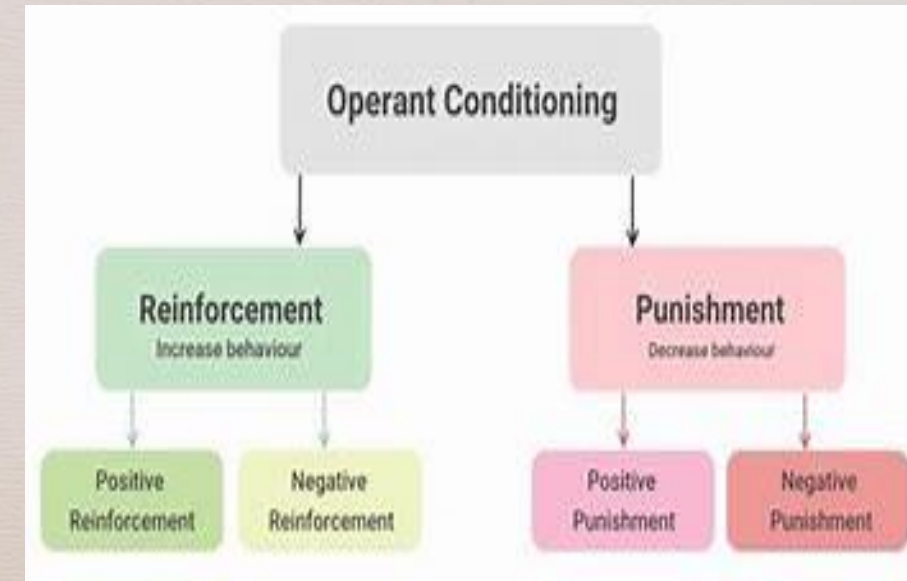
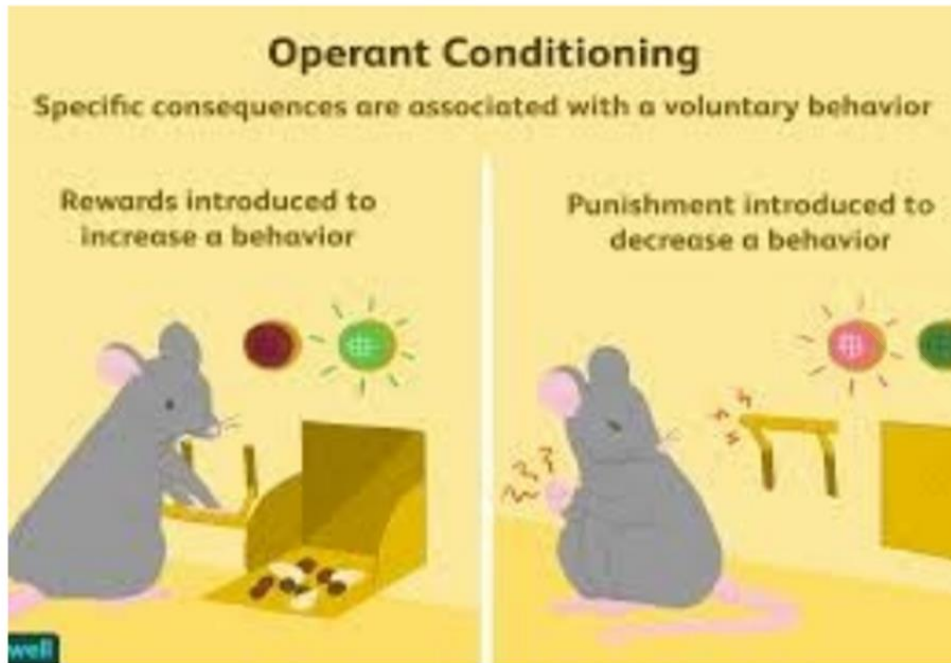


Figure 1.2. This illustration illustrates Operant Conditioning. The mouse pushes the lever and receives a food reward. Therefore, he will push the lever repeatedly in order to get the treat.

Learning: Operant Conditioning



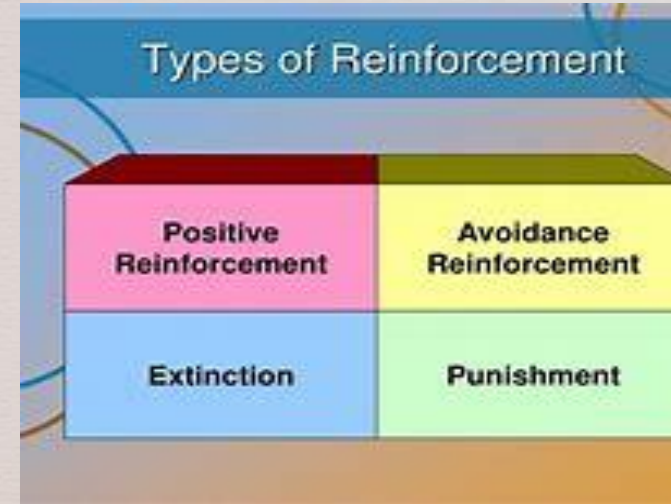
- A process through which organisms learn to repeat behaviours that yield positive outcomes or permit them to avoid or escape from negative outcomes



Key terms

- **Reinforcement:** The application or removal of a stimulus to increase the strength of a specific behaviour
- **Positive Reinforcement:** Stimuli that strengthen responses that precede them.

Example: praise and rewards



- **Negative reinforcement:** Stimuli that strengthen responses that permit the organism to avoid or escape from their presence.

- *Example:*

The penalty rules act as negative reinforcers to strengthen desirable behaviour.

Positive Punishment: A procedure by which the application of a stimulus decreases the strength of a behaviour.

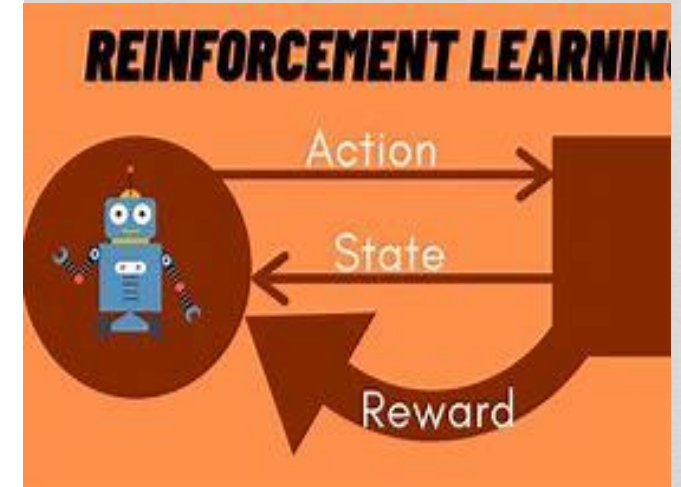
- *Example:*

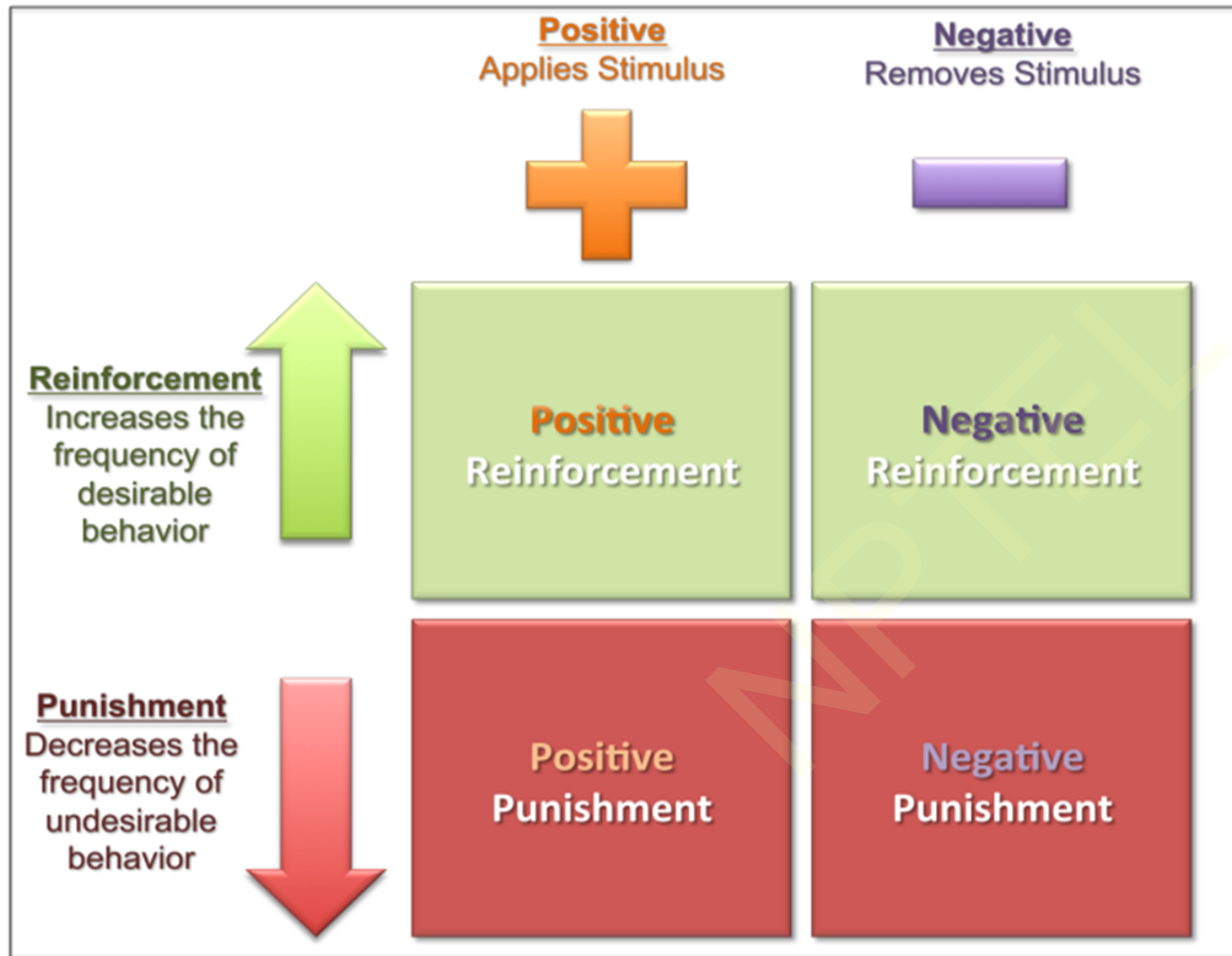
Disciplining (e.g. berating) a child for misbehaving

- **Negative Punishment:** When you remove something in order to decrease a behavior, this is called negative punishment.

- *Example:*

Putting a child in a time-out until they can decrease their aggressive behavior





How Reinforcement Schedules Work

Continuous Reinforcement vs. Partial reinforcement

Continuous Reinforcement



Desired behavior is reinforced every time it occurs



Most effective when teaching a new behavior

"SHAKE!"



Creates a strong association between behavior and response

verywell

Partial Reinforcement



Most effective once a behavior has been established



New behavior is less likely to disappear



Various partial reinforcement schedules available to suit individual needs





Schedules of reinforcement

- Rules determining when and how reinforcement will be delivered
 - **Continuous Reinforcement Schedule:** a schedule of reinforcement in which every occurrence of a particular behaviour is reinforced.
 - **Fixed-interval schedule:** A schedule of reinforcement in which a specific interval of time must elapse before a response will yield reinforcement.
 - **Variable-interval Schedule:** A schedule of reinforcement in which a variable amount of time must elapse before a response will yield reinforcement.
 - **Fixed-ratio Schedule:** A schedule of reinforcement in which reinforcement is delivered after a fixed number of responses have been performed.
 - **Variable-ratio schedule:** A schedule of reinforcement in which reinforcement is delivered after a variable number of responses have been performed.



Skinner believed the habits that each of us develops result from our unique operant learning experiences

Figure 1.3 Reinforcement and Punishment Comparison

	REINFORCEMENT (Behavior Increases)	REINFORCEMENT (Behavior Increases)
POSITIVE (Something is added.)	Positive Reinforcement Something is added to increase desired behavior. Ex: Smile and compliment student on good performance.	Positive Punishment Something is added to decrease undesired behavior. Ex: Give student detention for failing to follow the class rules.
NEGATIVE (Something is removed.)	Negative Reinforcement Something is removed to increase desired behavior. Ex: Give a free homework pass for turning in all assignments.	Negative Punishment Something is removed to decrease undesired behavior. Ex: Make student miss their time in recess for not following the class rules.

Extinction decreases the probability of a response by contingent withdrawal of a previously reinforced stimulus

➤ **Modeling , Shaping, and Cueing:**

➤ **Modeling** also known as observational learning

➤ **Shaping** is the process of gradually changing the quality of a response

➤ **Cueing** is providing a child with a verbal / non-verbal cue for the appropriateness of behaviour

➤ **Behaviour Modification:** has six basic components

1. **Specification of the desired outcome:** increased student participation in class discussions

2. **Development of a positive, nurturing environment**



3. Identification and use of appropriate reinforcers (**intrinsic and extrinsic rewards**)
4. **Reinforcement of behaviour patterns** develop until the student has established a pattern of success
5. **Reduction in the frequency** of rewards-a gradual decrease the amount of one-on-one review
6. **Evaluation and assessment** of the effectiveness of the approach: based on teacher expectations and student results

❑ **Criticisms:**

1. Oversimplifies the complexity of human behaviour
2. Ignores the internal psychological and mental processes
3. The principles of conditioning are not universal





❖ Educational Implications:

1. Students work for things that bring them **positive feelings**, and for approval from people they admire
2. **Behaviours can be learned, unlearned or relearned**
3. Ignoring an undesirable behaviour will go far toward eliminating it
4. **Organizing the time and duration of learning:**

studying intermittently or intensely;

Exmpl.- Students who learn in an overall manner by means of studying intensely just before an exam

are more successful at the exams;

- 5. The structure of the subject to be learnt: **learning, as a whole is superior to learning by parts**; learning by parts is more efficient in case of a lengthy topic



Lecture:7

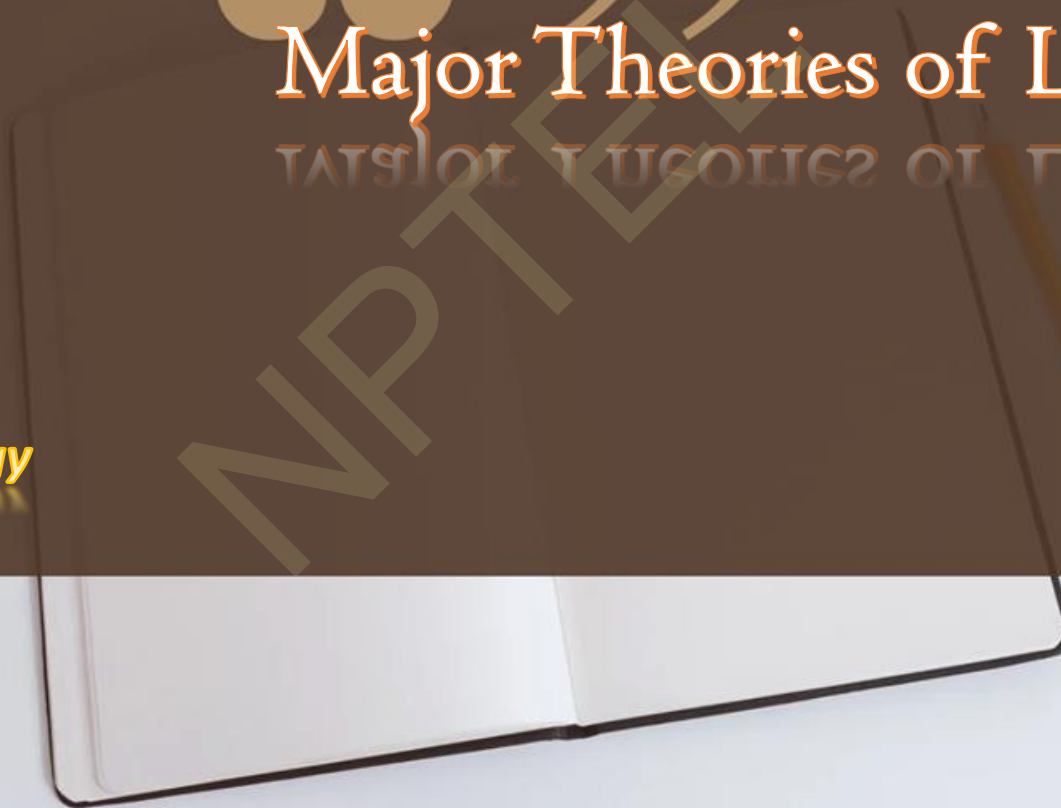


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6. **Participation**: attention, motivation and readiness

7. **Feedback**: Knowledge of the results for improvements.

❑ **Classical Conditioning** -is when an unconditioned stimulus and response is manipulated with a conditioned stimulus to create a conditioned response

❑ **Operant Conditioning**- is a controlled response with a reward/punishment system according to the behavior



- The **learner needs reinforcements** to keep interest
- **Stimuli are effective in controlling** behavior
- The **behaviors can be measured** to record learning success
- overall learning may fade away after a short time, can be forgotten if the exam is delayed;
- **Intermittent study is more efficient for** those who want to achieve a **permanent learning**

Operant Conditioning

Cognitive Perspective

- **Learned helplessness**

When people feel that they have no control over their situation, they may begin to behave in a helpless manner. This inaction can lead people to overlook opportunities for relief or change.

For example, a woman who feels shy in social situations may eventually begin to feel that there is nothing she can do to overcome her symptoms. This sense that her symptoms are out of her direct control may lead her to stop trying to engage herself in social situations, thus making her shyness even more pronounced.



Observational learning

(Bandura 1977)

Albert Bandura believed that direct reinforcement and punishment could not account for all learning that occurs. What about when children copy the behaviour of parents when no reinforcement is given?

Albert
Bandura

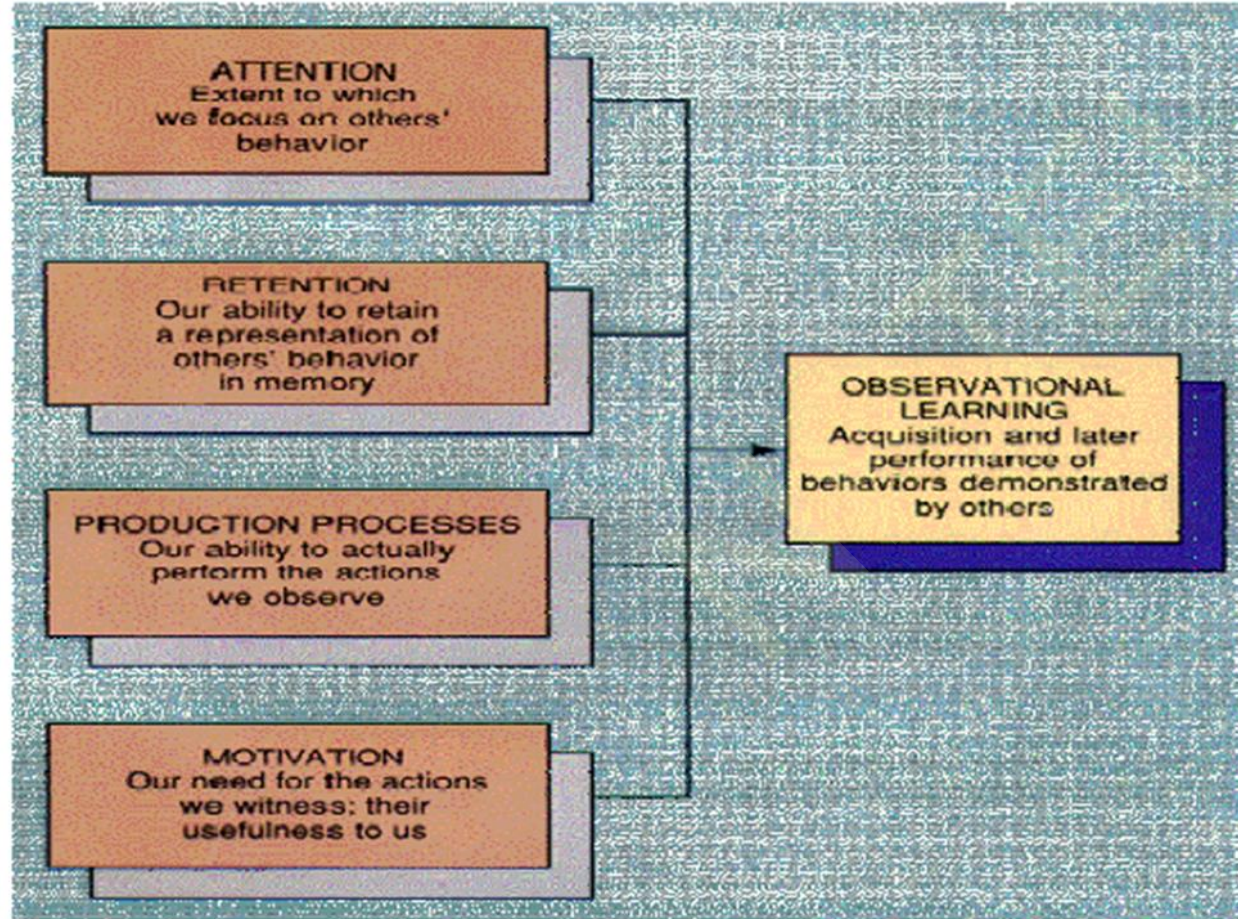


Observational learning: Basic Principles

- Must direct attention to specific model
- Must be able to remember what the person had said or done
- Need to be able to convert these memory representations into appropriate actions.
- Motivation is important. Only if the person finds the information useful, put them into action.



Key factors in observational learning



Learning

The process by which experience or practice results in a relatively permanent change in behavior or potential behavior



We are
here

Classical Conditioning

The type of learning in which a response naturally elicited by one stimulus becomes to be elicited by a different formally neutral stimulus

Pavlov and Watson

UCS, UCR, CS, CR

Operant Conditioning

The type of learning in which behaviors are emitted to earn rewards or avoid punishments

B.F. Skinner

Reinforcement and
Punishment

Social Cognitive Learning Theory

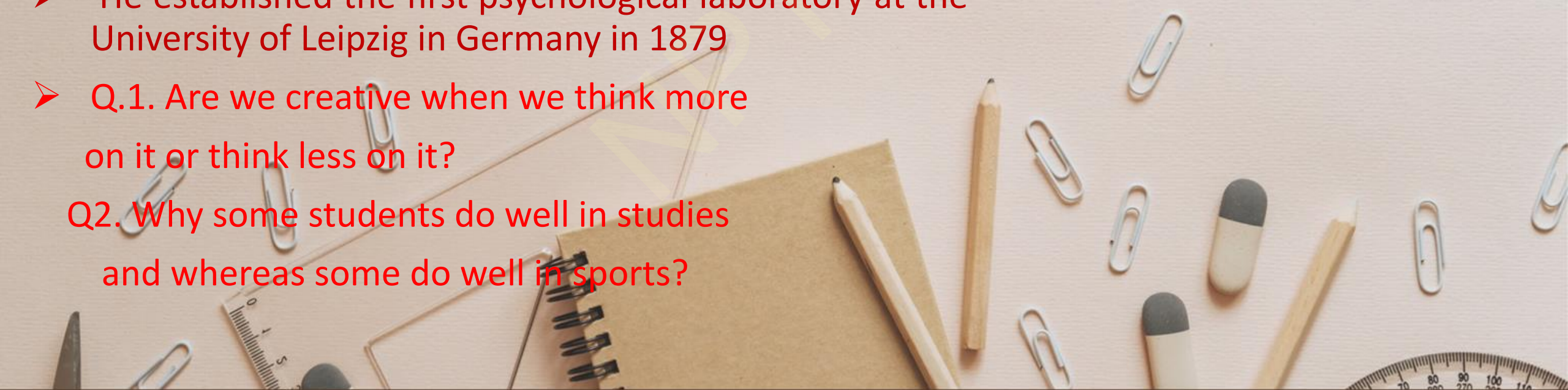
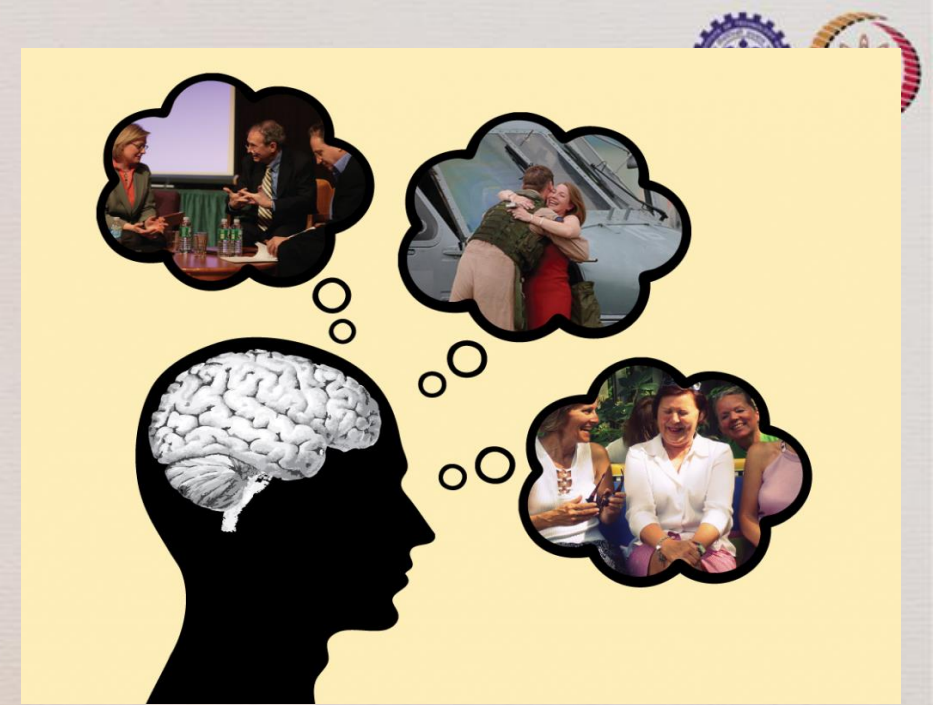
The type of learning in which behaviors are learned by observing a model

Albert Bandura

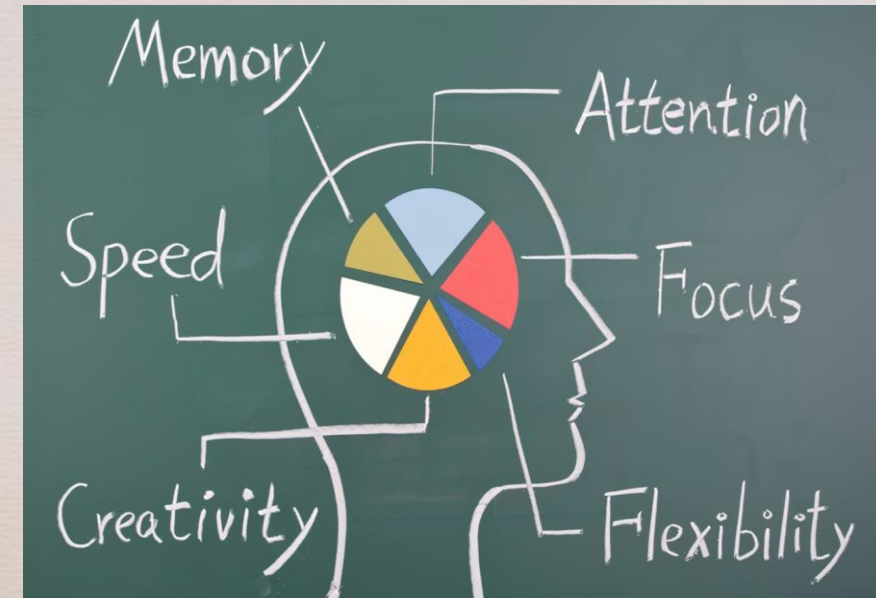
Modeling and Vicarious
Learning

Psychology

- The study of Human Behavior
- Derived from two Greek words: Psyche (mind/soul) & Logos (study)
- Founders: Socrates, Plato and Aristotle,
- The father of modern Psychology: Wilhelm Wundt
- He established the first psychological laboratory at the University of Leipzig in Germany in 1879
- Q.1. Are we creative when we think more on it or think less on it?
- Q2. Why some students do well in studies and whereas some do well in sports?



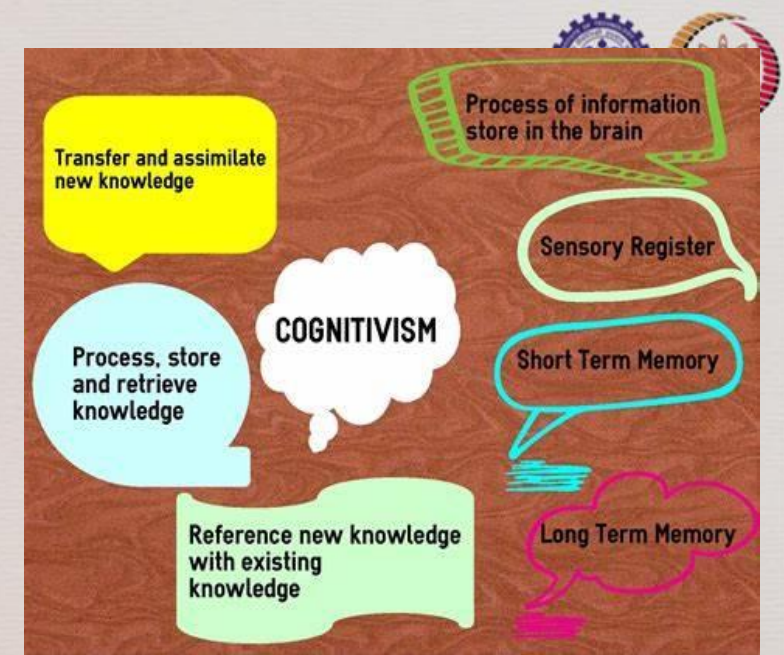
- **Learning** can be defined as an **observable change in the behavior** of a specific organism as a consequence of regularities in the environment of that organism.
- In order to say that learning has occurred, two conditions must be met:
- An **observable change in behavior** must occur during the lifetime of the organism.
 - The change in behavior must be **due to regularities** in the environment.
 - **Learning is thus seen as an effect**—that is, as an observable change in behavior that is attributed to an element in the environment (a regularity in the environment).
 - Learning takes place whenever **experience** or **practice results** in a relatively **permanent change in behavior** or behavioral potential (Gerrig & Zimbardo, 2008).



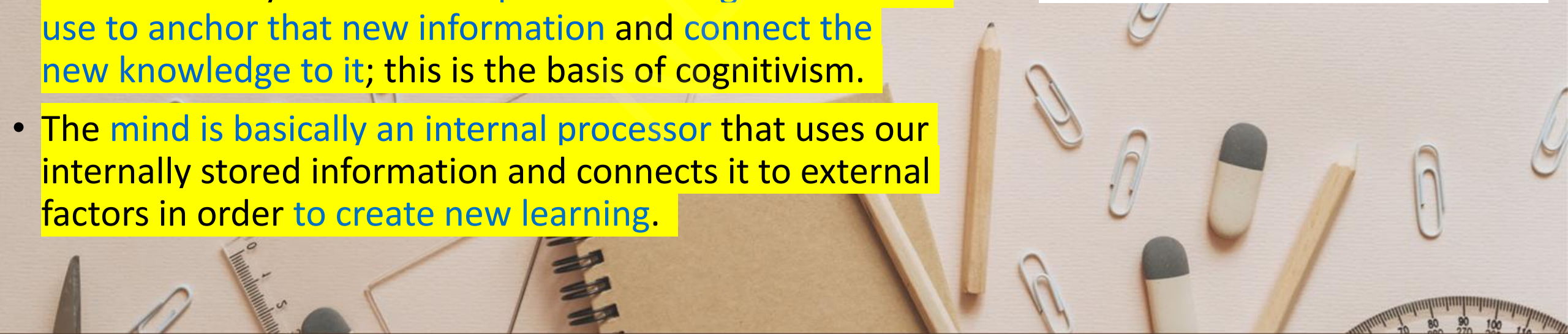
Cognitivism

- Cognitivism is a learning theory that focusses on how information is received, organized, stored and retrieved by the mind.
- It uses the mind as an information processor, like a computer.
- Therefore, cognitivism looks beyond observable behavior, viewing learning as internal mental processes.
- According to this perspective, learners are actively involved in the way they process information.
- Knowledge, memory, thinking, and problem solving are areas for development.
- Cognitivists objected to behaviorists because they felt that behaviorists thought learning was simply a reaction to a stimulus and ignored the idea that thinking plays an important role.
- Knowledge is an internal process rather than a product.

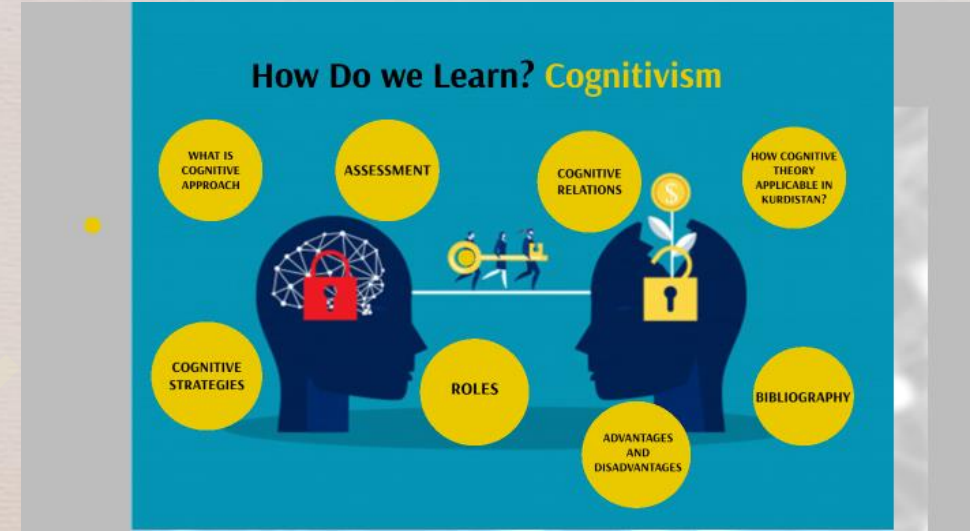
(J. Bruner)



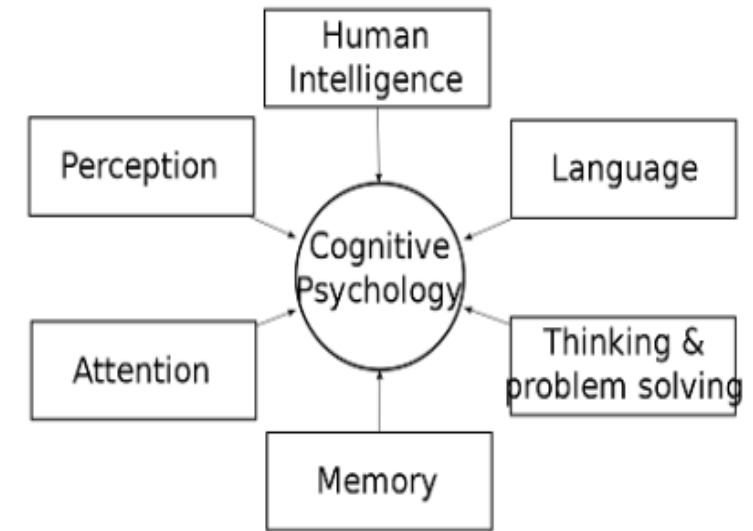
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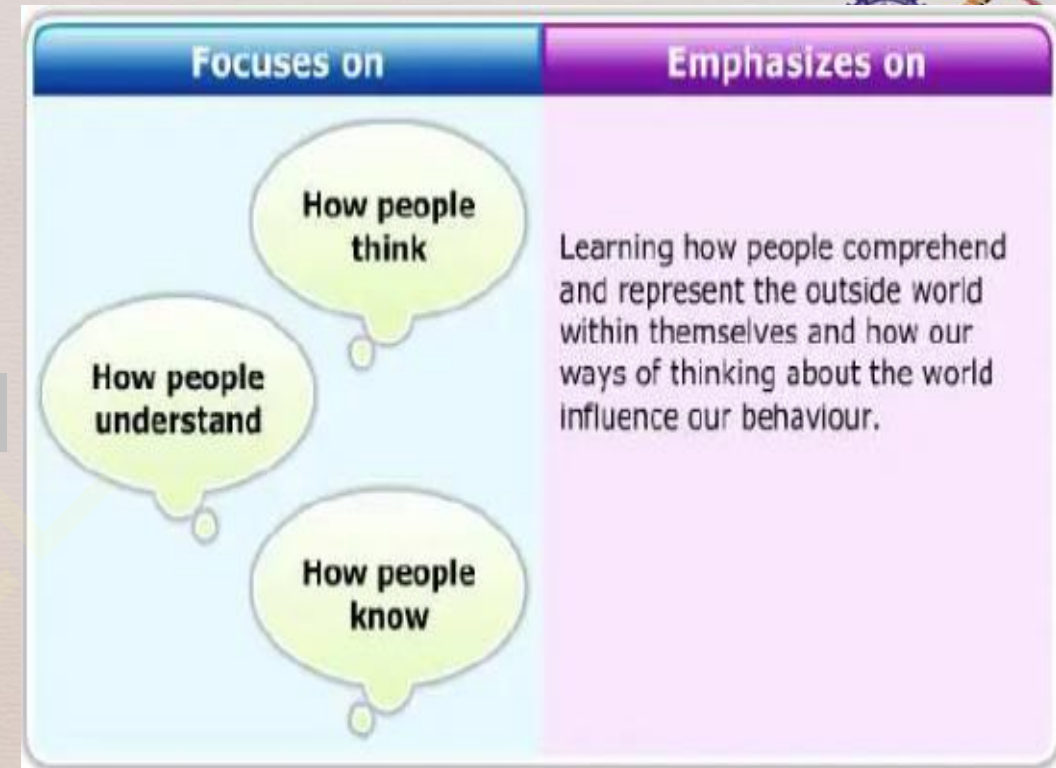
- Cognitivists believe that their theory is the **primary** foundation for **explaining how we learn things**.
- Cognitivism is viewed as the mainstream for all **research on learning designs**.
- Cognitivism is the theory that focuses on **how we receive, organize, store, and recall information in our minds**.
- Cognitivism theory of learning uses the concept that our **mind is like a computer** and explains **how it accesses schema** (file folders) and then uses the stored information to **create new learning** and **store it in a practical place**.
- If the **new information is stored properly and permanently**, then **learning has taken place**.



- ✓ Cognitivism holds that a person's **behavior is more dependent on insight into the relationships** that exist in a situation.
- ✓ The **whole is more than the sum of its parts**. They put pressure on the organization of observations for stimuli in the environment as well as on the factors that influence consideration.
- ✓ **Mental life encompasses cognitive, affective, conative symptoms** to some degree, namely **psychosomatics** that cannot be separated from each other.
- ✓ Thus, cognitive psychology not only explores the basis of typical **cognitive symptoms** but also from the **affective** (interpretation and consideration that accompanies the reaction of feelings), **conative** (will, passion, decisions).



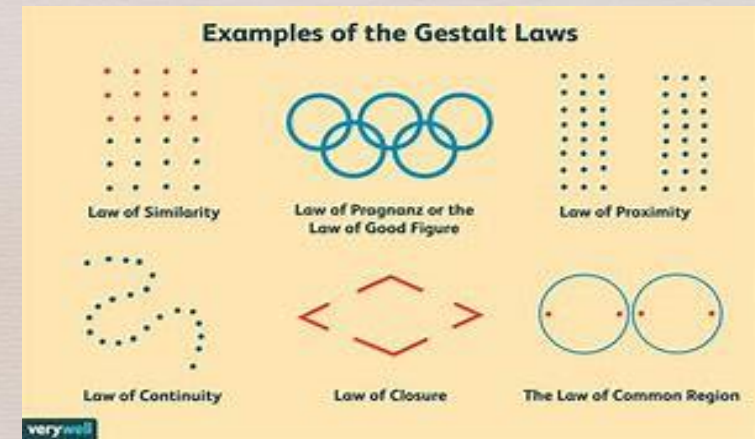
- The works of Edward C. Tolman, Jean Piaget, Lev Vygotsky, Jerome Bruner, and German Gestalt psychologists were instrumental in engendering the dramatic shift from behaviorism to cognitive theories.
- Tolman is usually considered a pioneer in initiating the cognitive movement; saw motivation as the key to transmuting expectations into behavior; often treated as a precursor of contemporary cognitive psychology.
- From a cognitive learning perspective, learning involves the transformation of information in the environment into knowledge that is stored in the mind.
- Learning occurs when new knowledge is acquired or existing knowledge is modified by experience.



• Gestalt Psychology

Gestalt is a perspective focuses on the belief that human consciousness cannot be broken down into its elements. This approach to psychology was founded on the concept of the *gestalt*, or **whole**. Gestalt psychologists led by Max Wertheimer (1880 - 1943), Wolfgang Kohler (1887 – 1967) and Kurt Koffka (1886 – 1941) have made substantial contributions to our understanding of perception. Gestaltists pointed out that perception has meaning only when it is seen as a whole.

The word Gestalt in German literally means “shape” or “figure”. Gestaltists performed many researches on perception and human learning. They believed learning is the result from good perception, which enable an individual to form correct concept in their mind. Later on they proposed the principles or law for perceptual organization. Henceforth, we will discuss these principles, which consisted of six principles, which are good form, figure or ground, similarity, proximity, closure, and continuity.





Lecture:8

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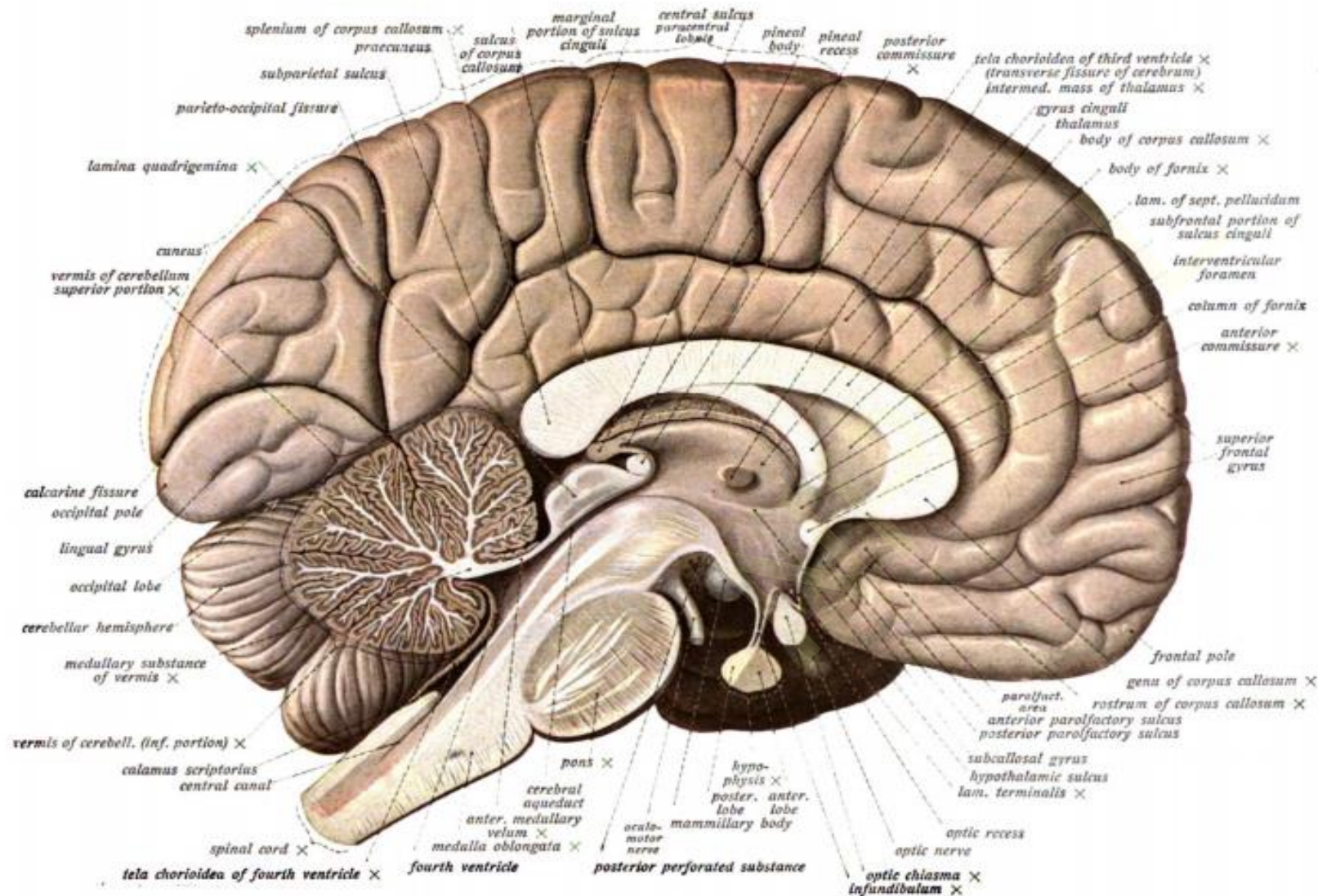
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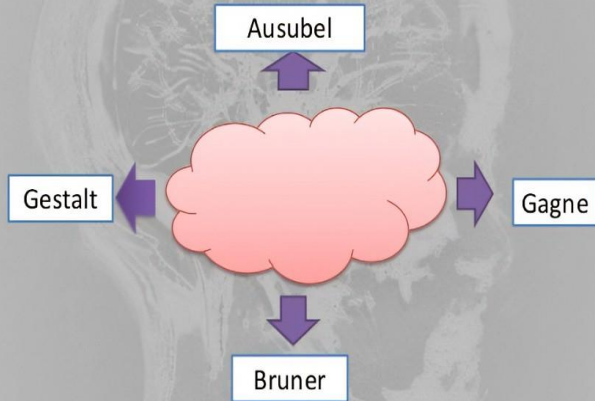
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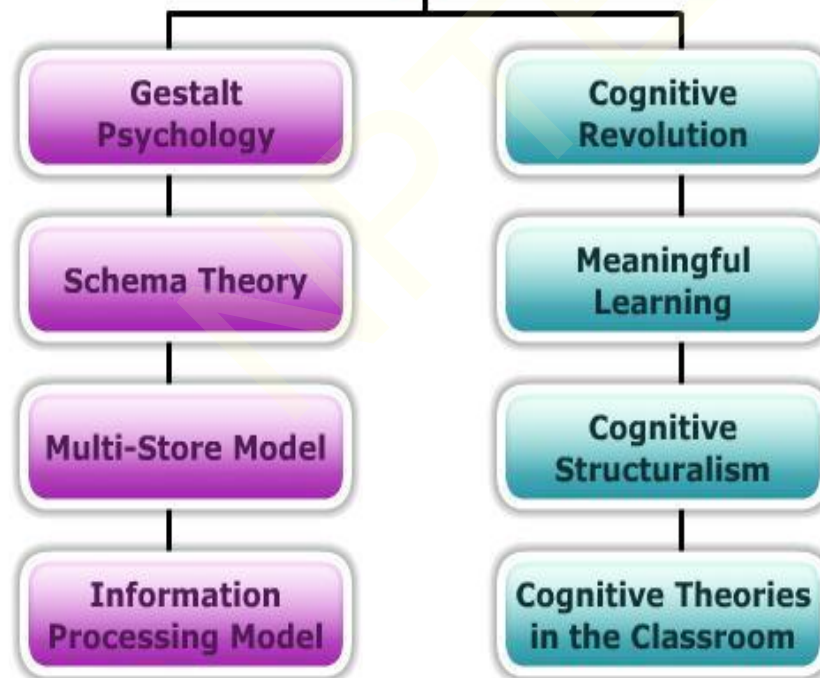
Biology and Behaviour



Cognitive learning models

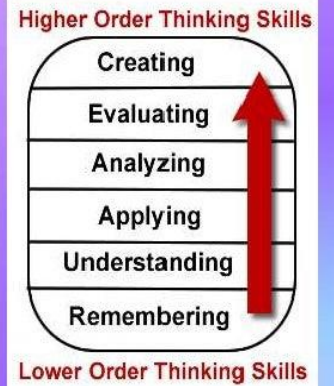


Cognitive Learning Theory

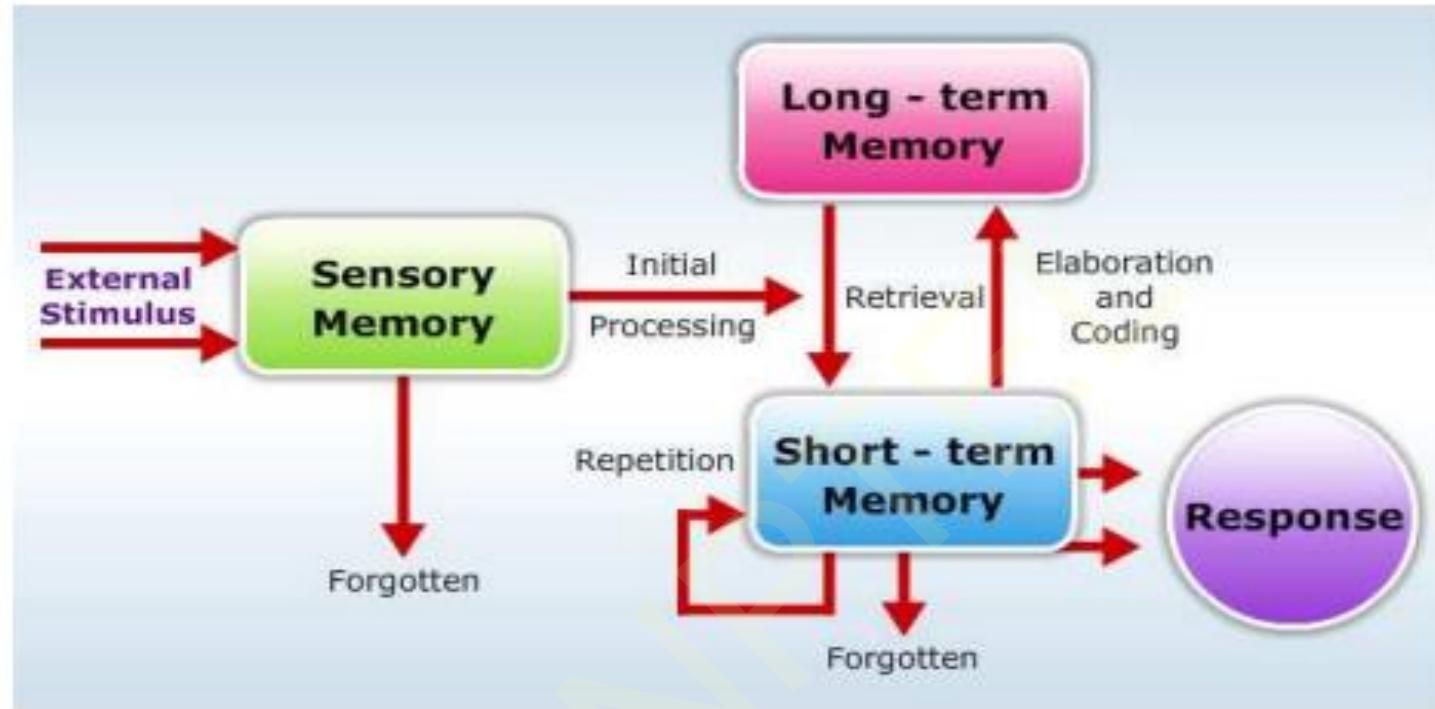


Benjamin Bloom (1913-1999)

- Educational psychologist who focused on students' learning domains
- Classified learning into three domains (cognitive, affective, and psychomotor)
- Created Bloom's Taxonomy which is made up of six levels that can be used to acquire knowledge on a topic (shown to the right)



The **information processing theory** offers detail descriptions of cognitive processes.



The Information Processing Theory approach to the study of cognitive development evolved out of the American experimental tradition in psychology. Information processing theorists proposed that like a computer, a human mind is a system that processes information through the application of logical rules and strategies. The mind has a limited capacity for the amount and nature of the information it can process.



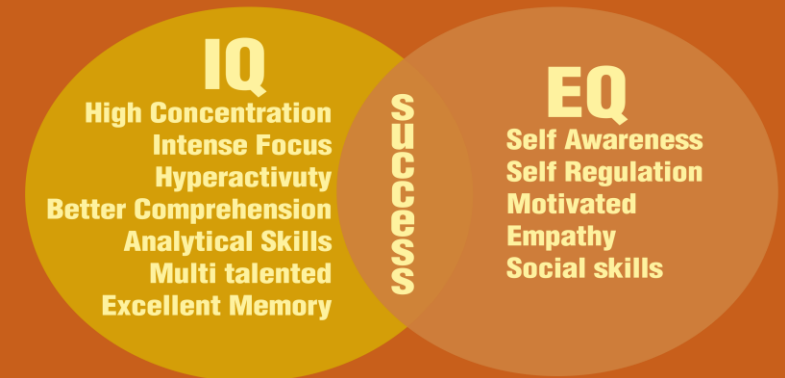
Intelligence

- ❑ Intelligence Individual's abilities to understand complex ideas to adapt effectively to the environment, to learn from experience, to engage in various forms of reasoning, to overcome obstacles by Careful thought
- ❑ "Intelligence is the global capacity of a person to act purposefully, to think rationally, and to deal effectively with his environment."-Wechsler
- Intelligence is typically measured as Intelligence Quotient (IQ)
- Standardized tests of intelligence usually measure so-called underlying cognitive ability: things like memory, speed of reactions and attention, as opposed to people's knowledge of facts.

Intelligence Quotient

- How is the Stanford-Binet scored?
 - IQ (intelligence quotient)

$$IQ = \frac{\text{mental age}}{\text{chronological age}} \times 100$$



Gardner's theory of multiple intelligence



Visual-Spatial

$$x^2 + y^2 = z^2$$



Linguistic-Verbal



Interpersonal



Intrapersonal



Logical-Mathematical



Musical



Bodily-Kinesthetic

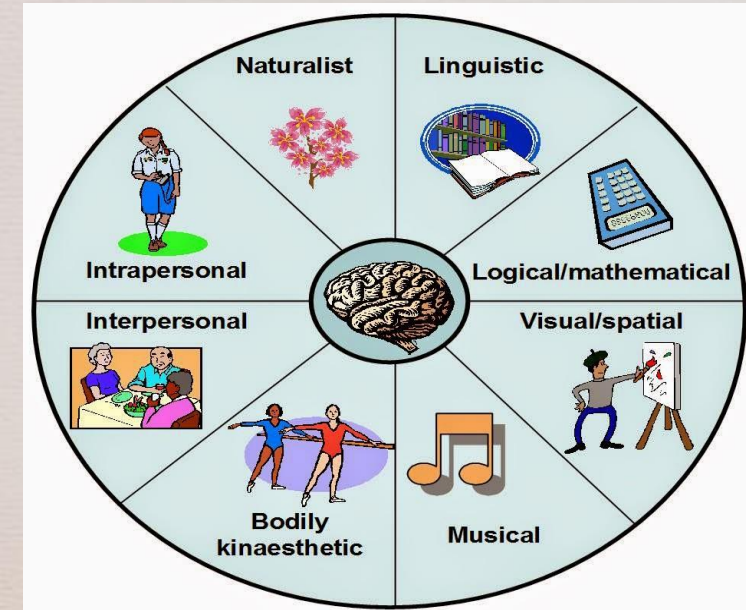


Naturalistic



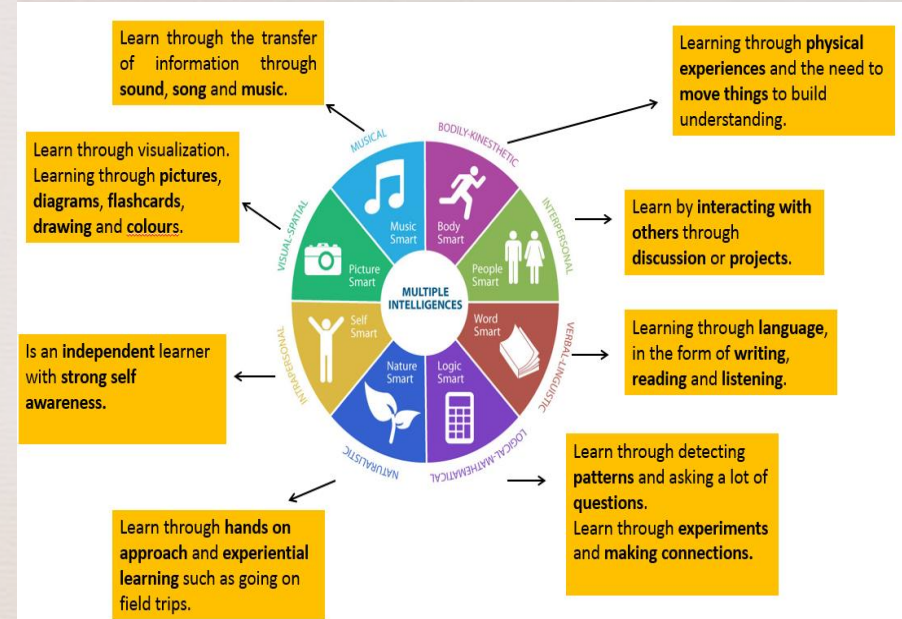


- **Howard Gardner** identified **seven intelligences** in his studies in psychology, human cognition, and human potential.
- These intelligences were named by Gardner as **linguistic intelligence**, **logical–mathematical intelligence**, **spatial intelligence**, **bodily–kinesthetic intelligence**, **musical intelligence**, **interpersonal intelligence**, and **intrapersonal intelligence**.
- **Linguistic intelligence** is the **capacity to use words effectively**, whether orally or in writing. It involves the mastery of spoken and written language to express oneself or remember things.
- **Logical–mathematical intelligence** is the **capacity to use numbers effectively**, detect patterns, think logically, reason deductively, and carry out mathematical operations.
- These two kinds of intelligences are typically the abilities that are **expected by the traditional school environments** to support and assess most **IQ measures** or **tests of achievement**.





- **Spatial intelligence** is the ability to perceive the visual–spatial world accurately and involves **sensitivity to color, line, shape**, form, space, and the potential for recognizing and **manipulating the patterns of spaces**.
- **Bodily–Kinesthetic intelligence** includes an **expertise in using one's whole body or parts of the body to express ideas** and feelings; and solve problems or create products.
- **Musical intelligence** is the capacity to **perceive, discriminate, transform, and express musical forms**, and use them **for performance or composition**.
- **Interpersonal intelligence** is the **capacity to perceive and make distinctions in the moods, intentions, motivations, and feelings of other people**.
- The last intelligence is the **intrapersonal intelligence**, and this intelligence is about **self-knowledge** and the **ability to act adaptively** on the basis of that knowledge. (Gardner, 1991)



- Later on Gardner also proposed that one more intelligence, **naturalistic intelligence**, met the criteria for identification as an intelligence as well.

- **Naturalistic intelligence** involves high expertise in **recognition and classification of the numerous species** the flora and fauna—of the environment.

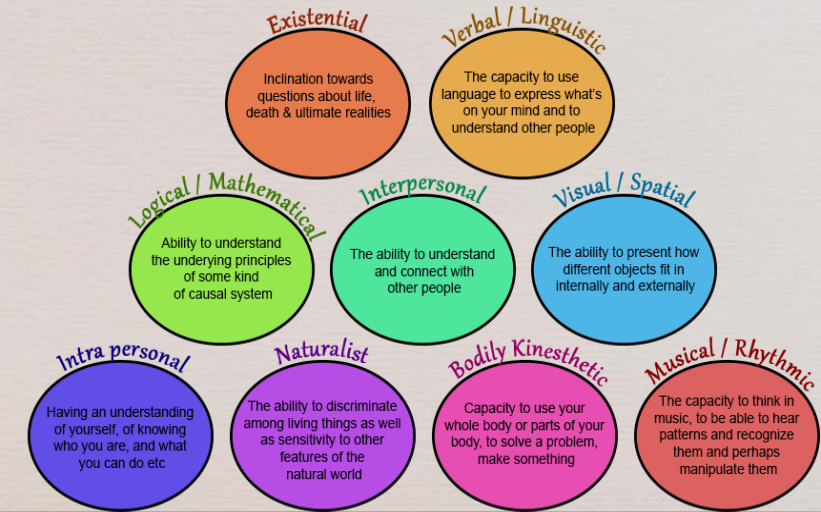
- More recently, Gardner has added an additional intelligence, the **existential intelligence**.

- He defines this **intelligence** as-

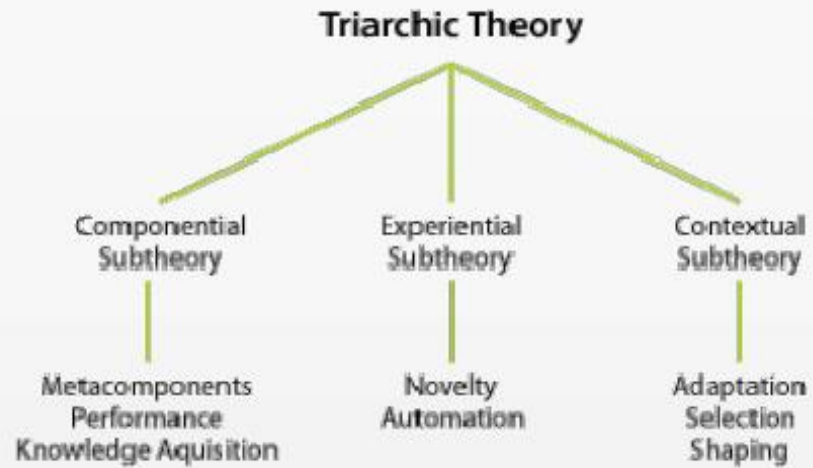
“The capacity to locate oneself with respect to the furthest reaches of the cosmos—the infinite and the infinitesimal—and the related capacity to locate oneself with respect to such existential features of the human condition as the significance of life, the meaning of death, the ultimate fate of the physical and the psychological worlds, and such profound experiences as love of another person or total immersion in a work of art” (**Gardner,1999**).



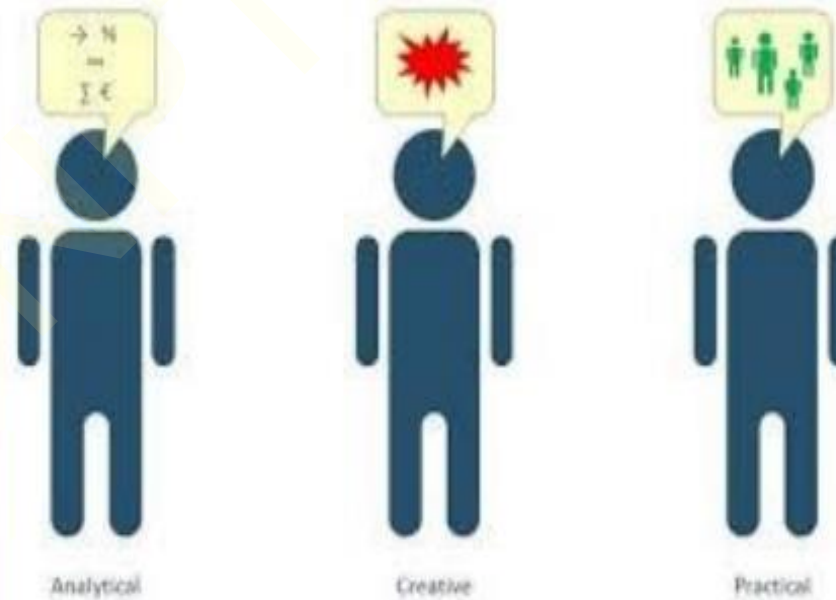
Multiple Intelligences



Sternberg's Triarchic theory of Intelligence



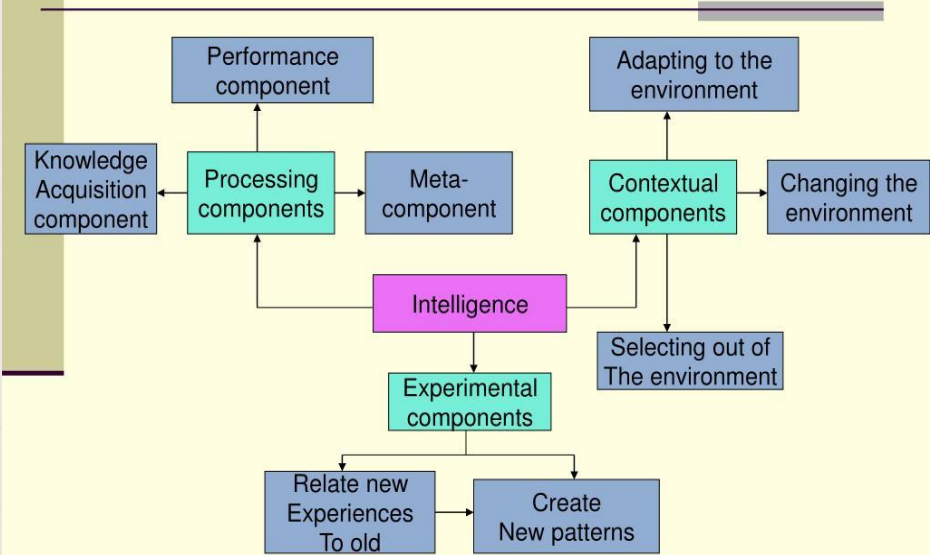
A theory suggests that there are three basic forms of intelligence:





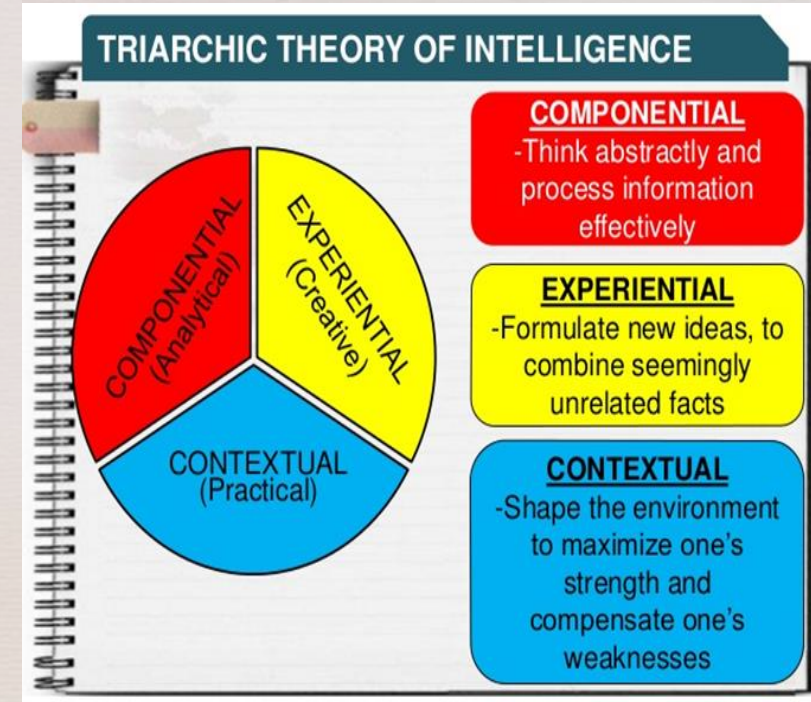
- **Robert J. Sternberg**, a well-known psychologist whose research often focuses on human intelligence and creativity.
- Sternberg proposed his theory in **1985** as an **alternative to the idea of the general intelligence factor**.
- The general intelligence factor, also known as *g*, is what intelligence tests typically measure; refers only to “academic intelligence.”
- Sternberg argued that **practical intelligence—a person’s ability to react and adapt to the world around them—as well as creativity are equally important** when measuring an individual's **overall intelligence**.
- He also argued that **intelligence isn’t fixed**, but rather **comprises a set of abilities** that can be developed.
- Sternberg's assertions led to the creation of his theory.

Sternberg's triarchic model of intelligence



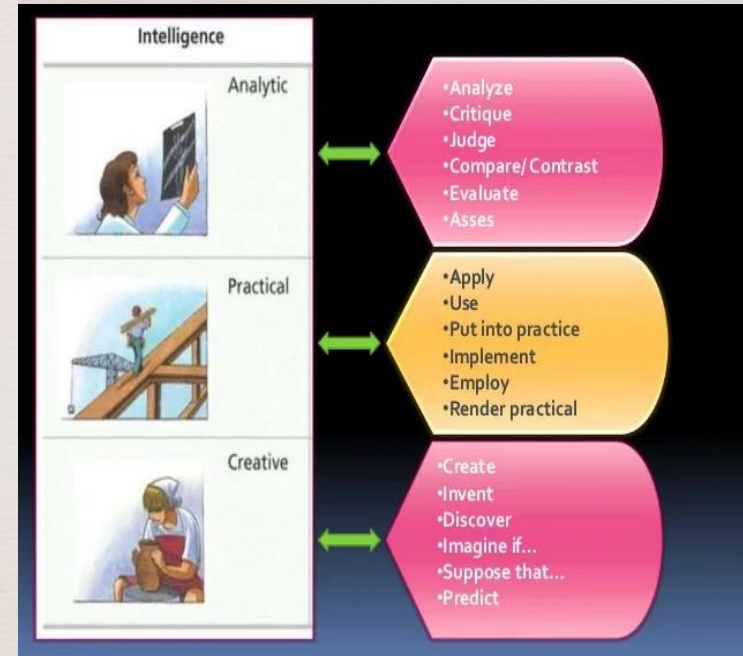


- The **Triarchic theory of intelligence** proposes that there are **three distinct types of intelligence: practical, creative, and analytical**.
- **Sternberg broke** his theory down into the following **three sub-theories**:
- **Contextual sub-theory**: The contextual sub-theory says that intelligence is intertwined with the individual's environment.
- Thus, **intelligence is based on the way one functions** in their everyday circumstances, **including one's ability to**
 - a) adapt to one's environment,
 - b) select the best environment for oneself, or
 - c) shape the environment to better fit one's needs and desires.
- **Experiential sub-theory**: The experiential sub-theory proposes that **there is a continuum of experience from novel to automation** to which intelligence can be applied.
- It's **at the extremes of this continuum that intelligence is best demonstrated**.





- At the **novel end of the spectrum**, an **individual is confronted with an unfamiliar task or situation** and must come up with a way to deal with it.
- At the **automation end of the spectrum**, **one has become familiar with a given task or situation** and can now handle it with minimal thought.
- **Componential sub-theory**: The componential theory outlines the various mechanisms that result in intelligence.
- This sub-theory is comprised of **three kinds of mental processes** or components:
- **Meta-components** enable us to monitor, control, and evaluate our mental processing, so that we can make decisions, solve problems, and create plans.
- **Performance components** are what enable us to take action on the plans and decisions arrived at by the meta-components.
- **Knowledge-acquisition components** enable us to learn new information that will help us carry out our plans.



- **Kinds of Intelligence-**
- **Practical intelligence:** Sternberg called **one's ability to successfully interact with the everyday world** practical intelligence.
- Practical intelligence is related to the **contextual sub-theory**.
- **Practically intelligent people are especially adept at behaving in successful ways** in their external environment.
- **Creative intelligence:** The experiential sub-theory is related to **creative intelligence**, which is **one's ability to use existing knowledge to create new ways to handle new problems** or cope in new situations.
- **Analytical intelligence:** The componential sub-theory is **related to** analytical intelligence, which is essentially academic intelligence.
- **Analytical intelligence is used to solve problems** and is the kind of intelligence that is measured by a **standard IQ test**

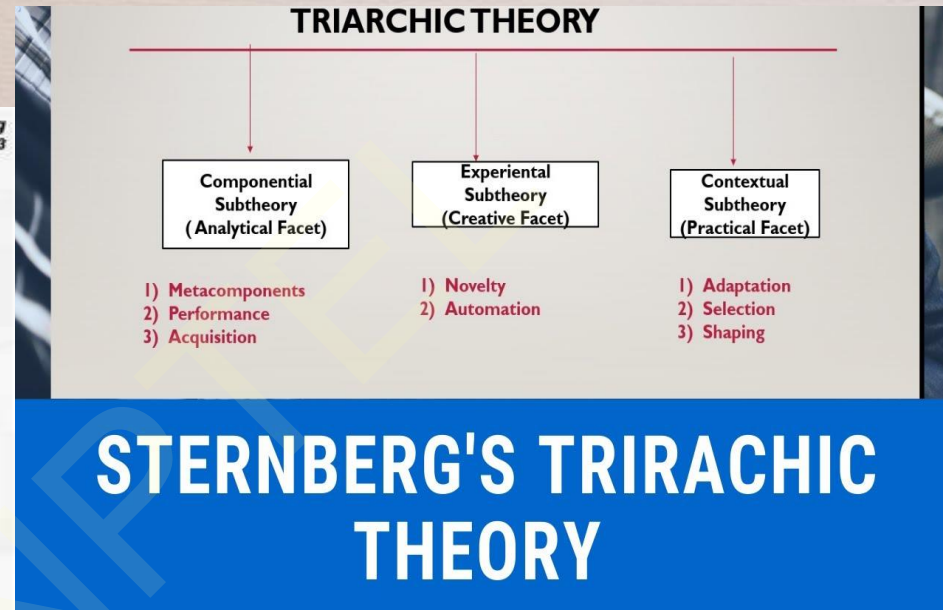
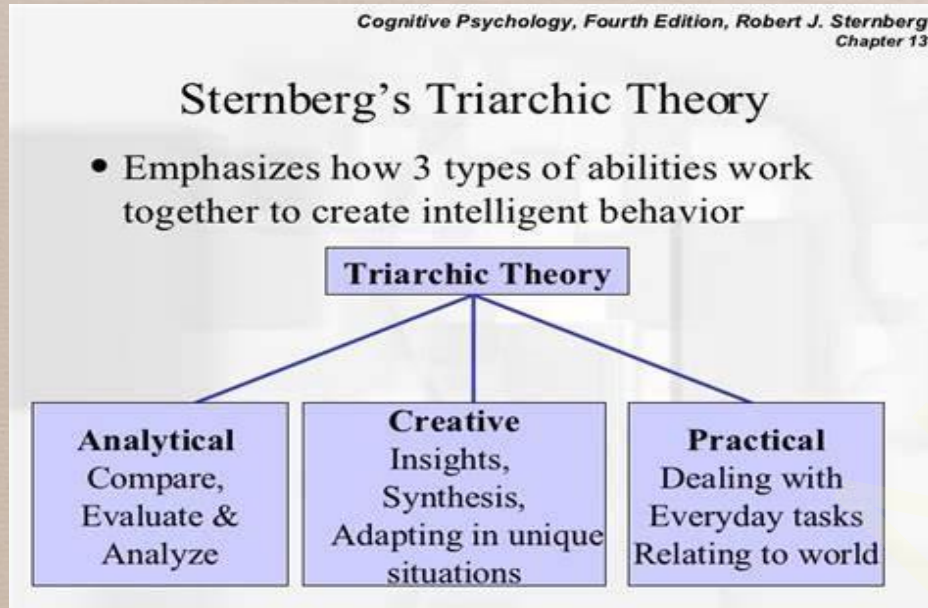
Triarchic Theory (Sternberg)



- Most commonly accepted theory today.
- **Three types of intelligence**
 1. **Analytical** - ability to analyze and evaluate ideas, solve problems and make decisions.
 - **Example:**
 2. **Creative** - going beyond what is given to generate novel and interesting ideas
 - **Example:**
 3. **Practical** - ability that individuals use to find the best fit between themselves and the demands of the environment.
 - **Example:**



Sternberg observed that **all three kinds of intelligence are necessary for successful intelligence**, which refers to **the ability to be successful in life based on one's abilities, personal desires, and environment**.





Lecture:9



Major Theories of Learning

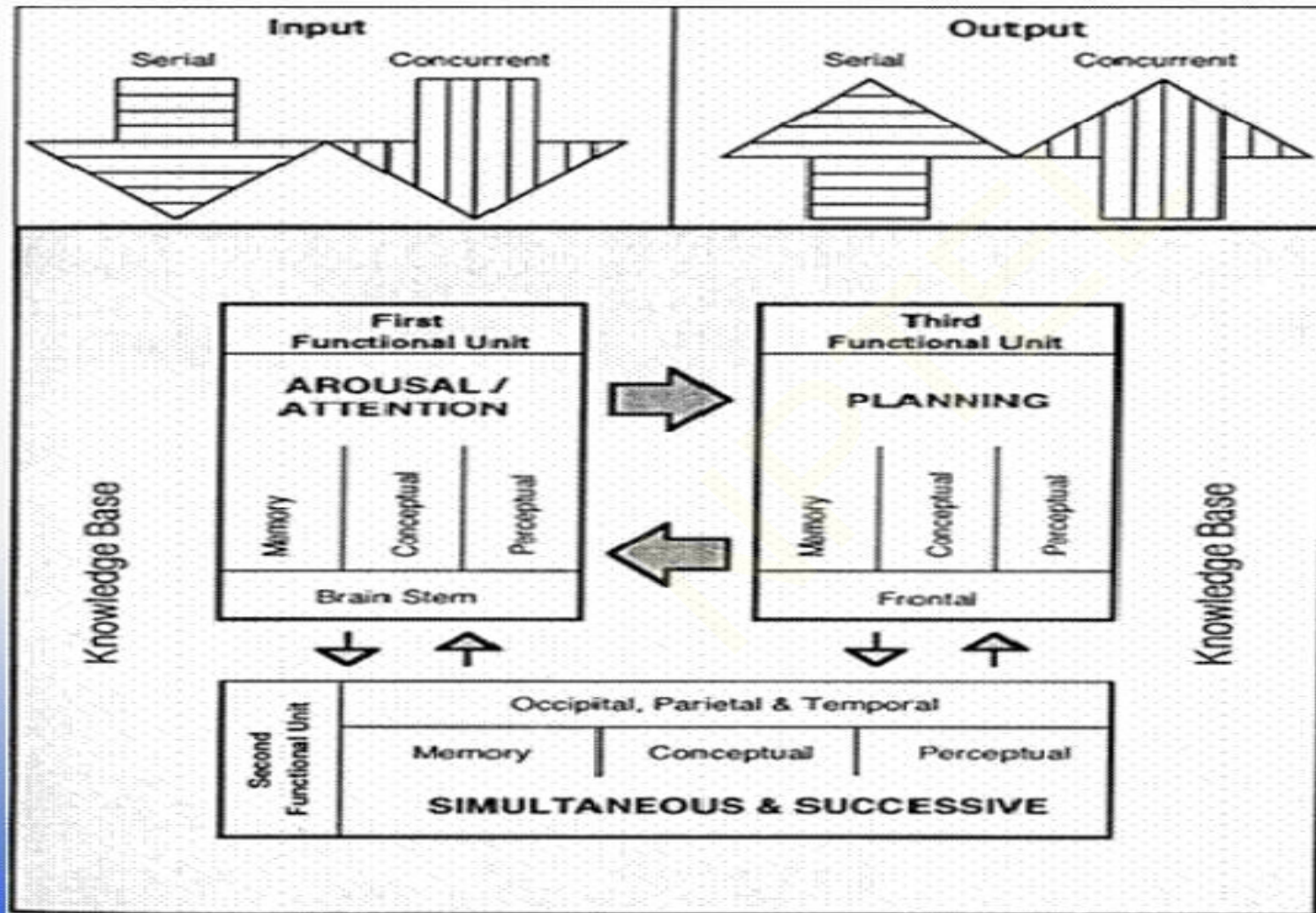
TATSLOI TUCOLIG2 OL TCSLUIUD

Prof. Atasi Mohanty
Indian Institute of Technology
Kharagpur

NPTEL

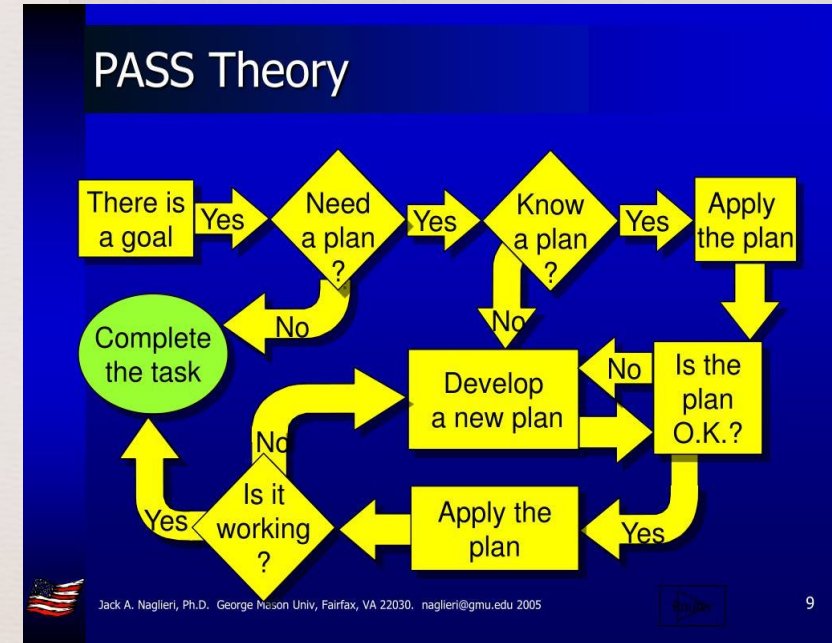
PASS Model of Intelligence

(Planning, Attention, Simultaneous, and Successive (PASS)-Das, Naglieri, & Kirby (1994))



• PASS Theory of Intelligence

- The **Planning, Attention-Arousal, Simultaneous and Successive (PASS)** theory of intelligence, was developed by Das, Nagliery and Kirby (1994).
- PASS theory is useful to link PASS processes to the brain.
- This theory has taken **its inspiration from the Pioneering** neuropsychological researches of **Alexander Luria**.
- Luria described **human cognitive processes** within the framework of **three functional units**.
- The function of the **first unit is cortical arousal and attention**, the **second unit codes information using simultaneous and successive processes** and the **third unit provides for planning, self-monitoring**, and structuring of cognitive activities.
- There are **four main components of the theory**, viz, Planning, Attention-Arousal, Simultaneous, and Successive (PASS) processing systems.

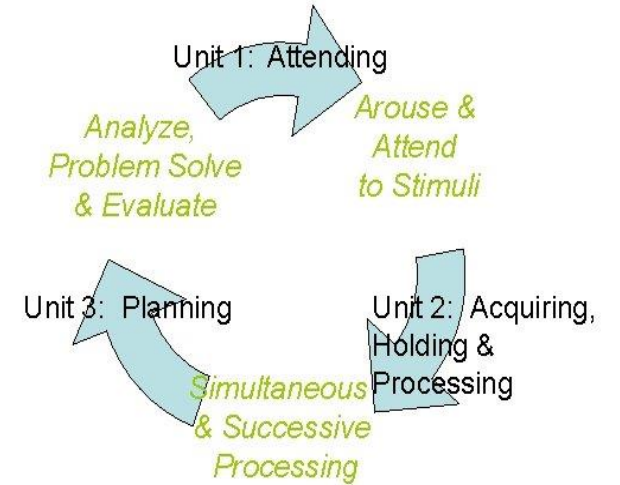




- **Planning** is a **frontal lobe function**. It is, more specifically, associated with the **prefrontal cortex** and one of the **main abilities** that **distinguishes humans from other primates**.
- The **prefrontal cortex** plays a central role in forming **goals and objectives** and then in **devising plans of action** required to attain these goals.
- It selects the **cognitive skills required to implement the plans**, coordinates these skills, and applies them in a correct order.
- Finally, the **prefrontal cortex is responsible for evaluating our actions as success or failure** relative to our intentions.
- **Planning**, helps us achieve through the selection or **development of plans or strategies needed to complete tasks** for which a solution is needed and is critical to all activities where the child or adult has to determine how to solve a problem.

Theories of Intelligence

The PASS Model



Beltre-Smith et al.
Mental Retardation, Seventh Edition

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- This includes **generation, evaluation, and execution** of a **plan** as well as **self-monitoring** and **impulse control**.
- Thus, **Planning** allows for the **solution of problems, control of attention, simultaneous, and successive processes**, as well as **selective utilization of knowledge and skills**.
- **Attention** is a mental process that is closely related to the **orienting response**.
- The base of the **brain allows** the organism **to direct focused selective attention toward a stimulus** over time and **resist loss of attention** to other stimuli.

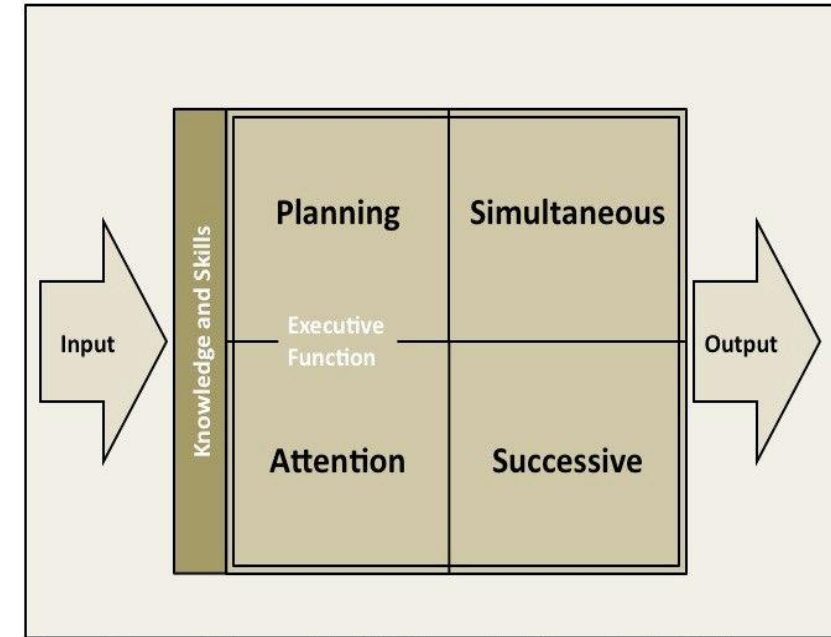
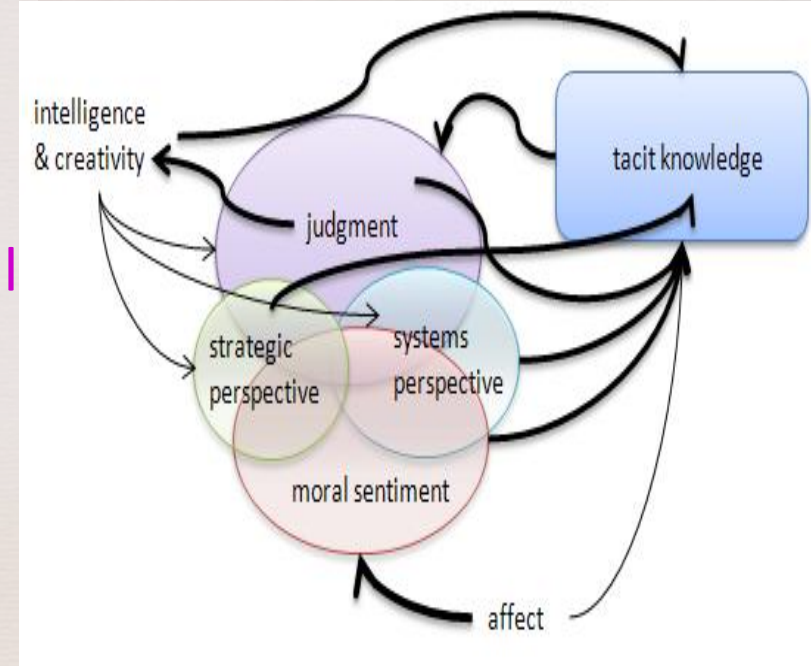


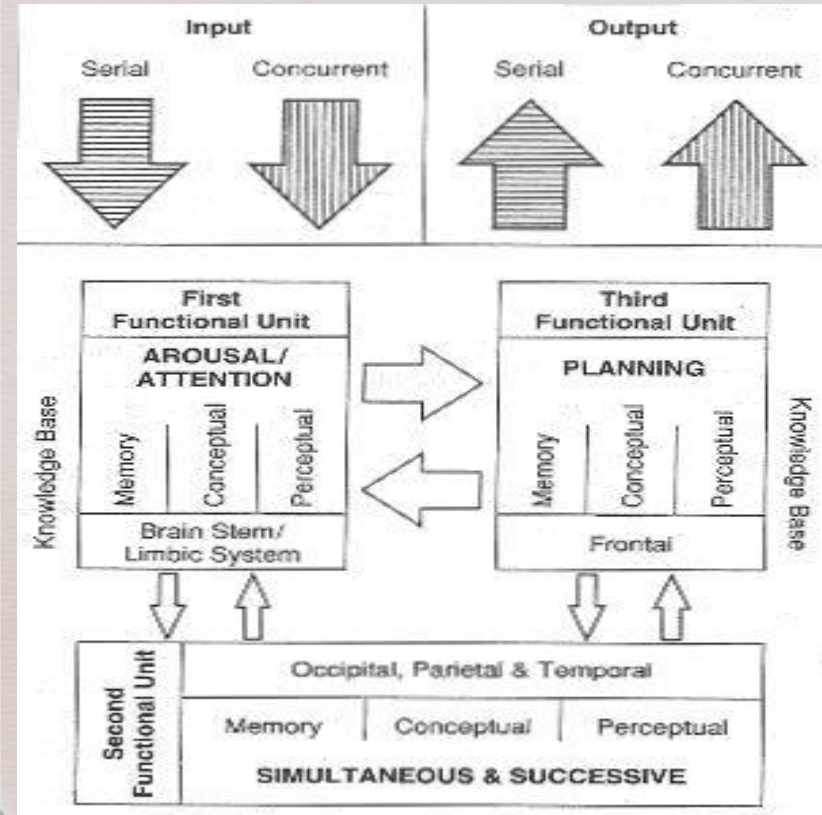
Figure 1. Functional Diagram of the PASS Theory of Intelligence. Naglieri 2011

- The **longer attention** is required the more the activity can be one that demands **vigilance**.
- **Attention** is controlled by **intentions** and **goals** and involves **knowledge** and **skills** as well as the other PASS processes.
- **Knowledge base** is closely associated with the PASS model and **all processes are found within this framework**.
- Knowledge base of the PASS model represents **all information obtained from the cultural and social background** and throws light on various parts of behavior.
- **Simultaneous Processing** is essential for **organization of information into groups or a coherent whole**.
- The **parietal-occipital-temporal brain** regions provide **a critical ability to see patterns as interrelated elements**.



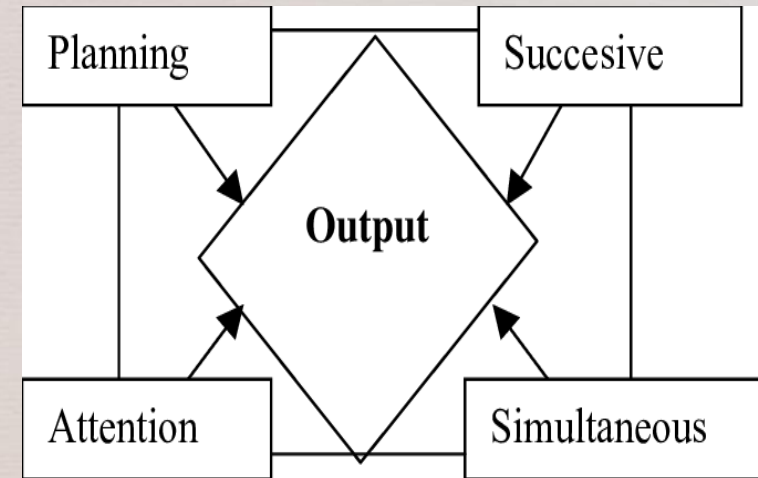


- **Planning** processes provide for the **programming, regulation and verification of behavior** and are responsible for behaviors, such as asking questions, **problem solving**, and the **capacity for self-monitoring**.
- **Successive Processing** is involved with the **use of stimuli arranged in a specific serial order**.
- Whenever **information** must be remembered or **completed in a specific order successive processing** will be involved.
- Importantly, however, the information must not be able to be organized into a pattern (like the number 9933811 organized into 99-33-8-11) but instead **each element can only be related to those that precede it**.
- The PASS theory is an alternative to approaches to intelligence that have traditionally included verbal, nonverbal, and quantitative tests.



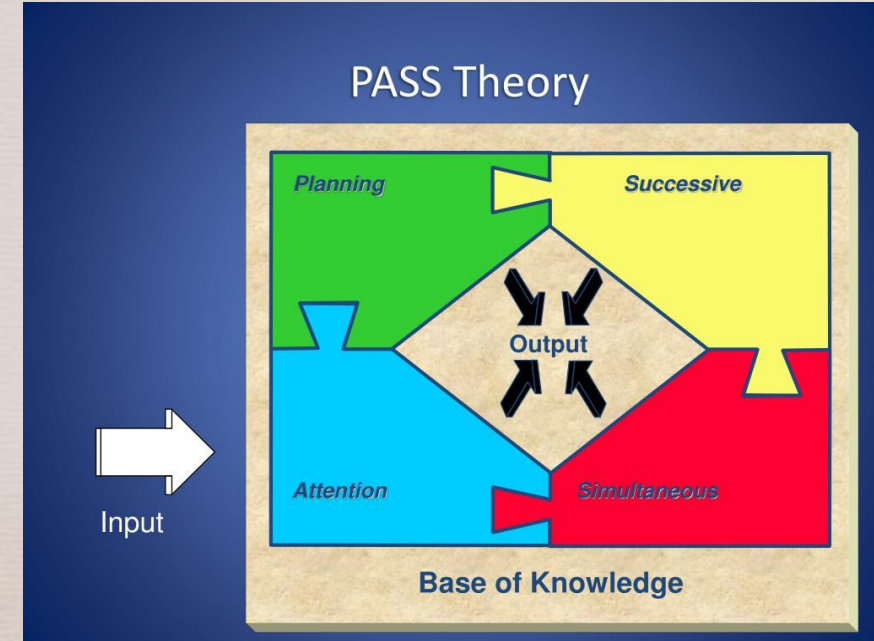


- PASS theory puts emphasis on basic psychological processes and precludes **verbal achievement** like tests such as vocabulary.
- The **functions of the brain** are considered the building blocks of ability conceptualized within a cognitive processing framework.
- PASS theory **applications** are spread over **developmental** and **educational psychology**.
- For the children in the age group of 8 to 17 years, **Ability/achievement, Ability/reading and ability/ mathematics normative taxonomies** for **reading** and **mathematics** have been developed from the **PASS theory** scales of the **CAS** (Cognitive Assessment System).



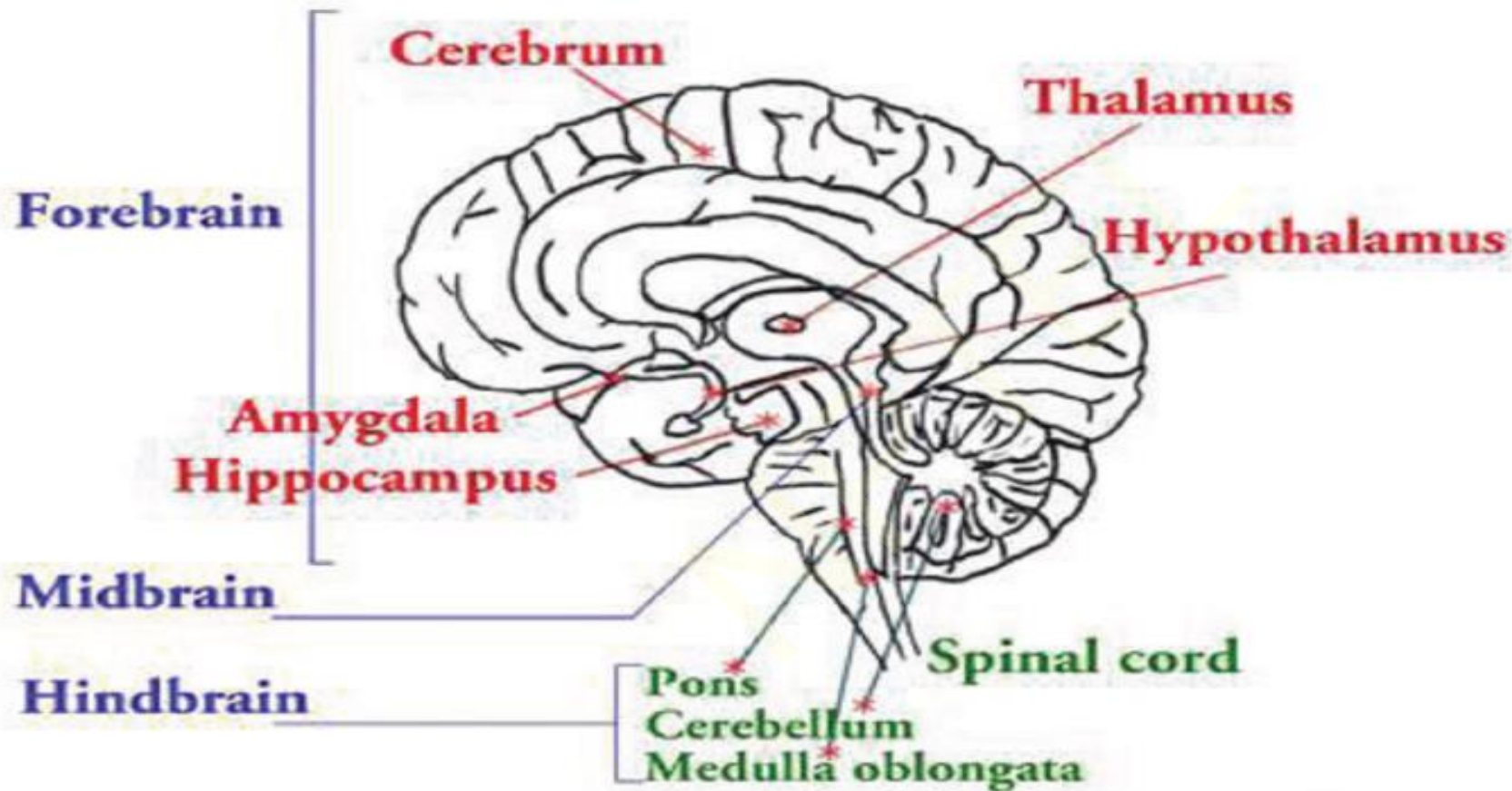
Salient features of PASS theory:

- It tests intelligence as a set of cognitive processes.
- The PASS theory of intelligence examines what the major processes are and guides us in the remediation of cognitive processing difficulties.
- Cognition is a dynamic mechanism that works inside the setting of the individual's knowledge base; it is also liable to developmental varieties.
- Cognitive evaluation for planning, attention, simultaneous, and successive processing of individuals can be done through CAS test battery based on PASS theory.



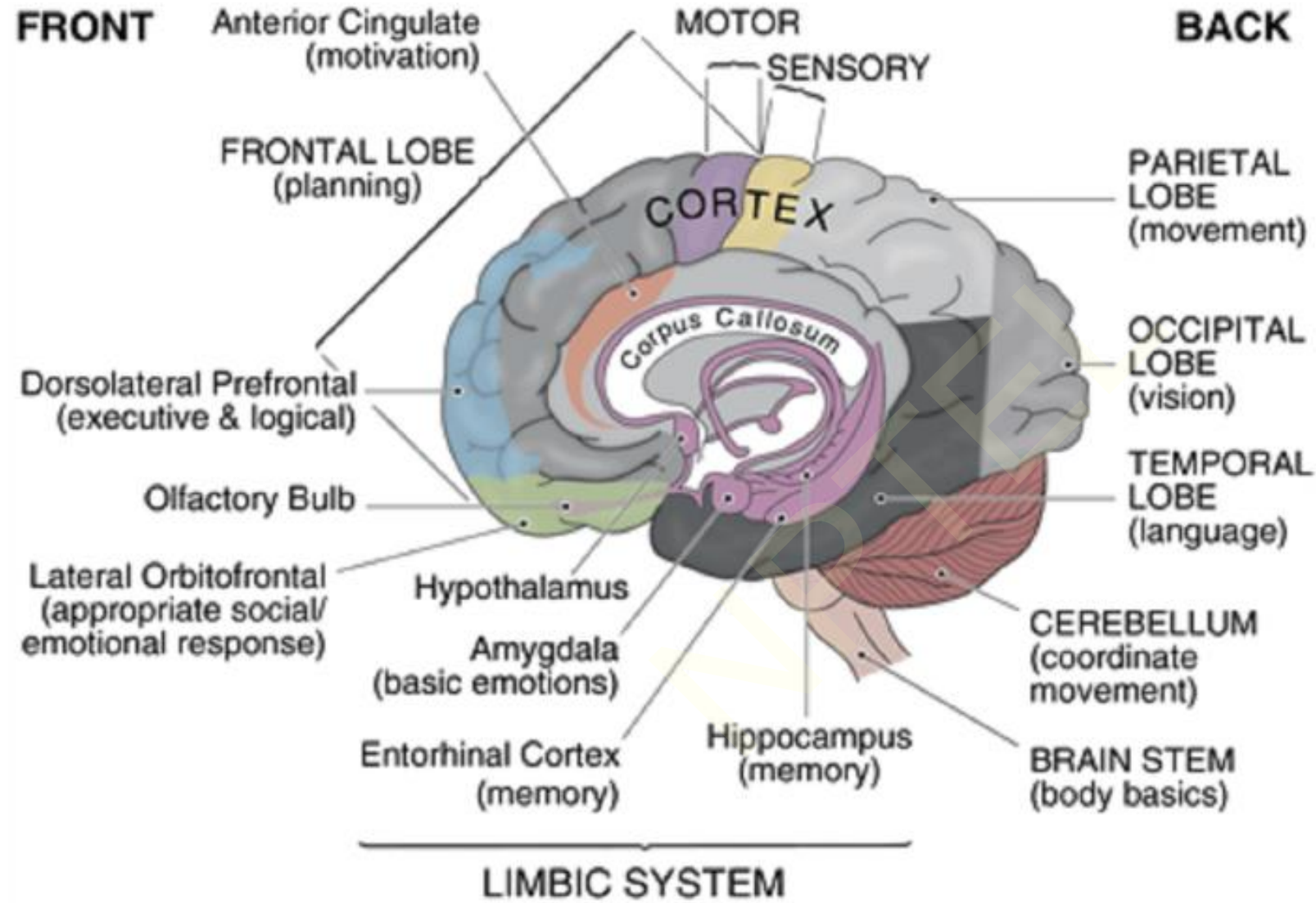
Cognitivism

BRAIN-BASED LEARNING



Graph 1: parts of forebrain, midbrain and hindbrain

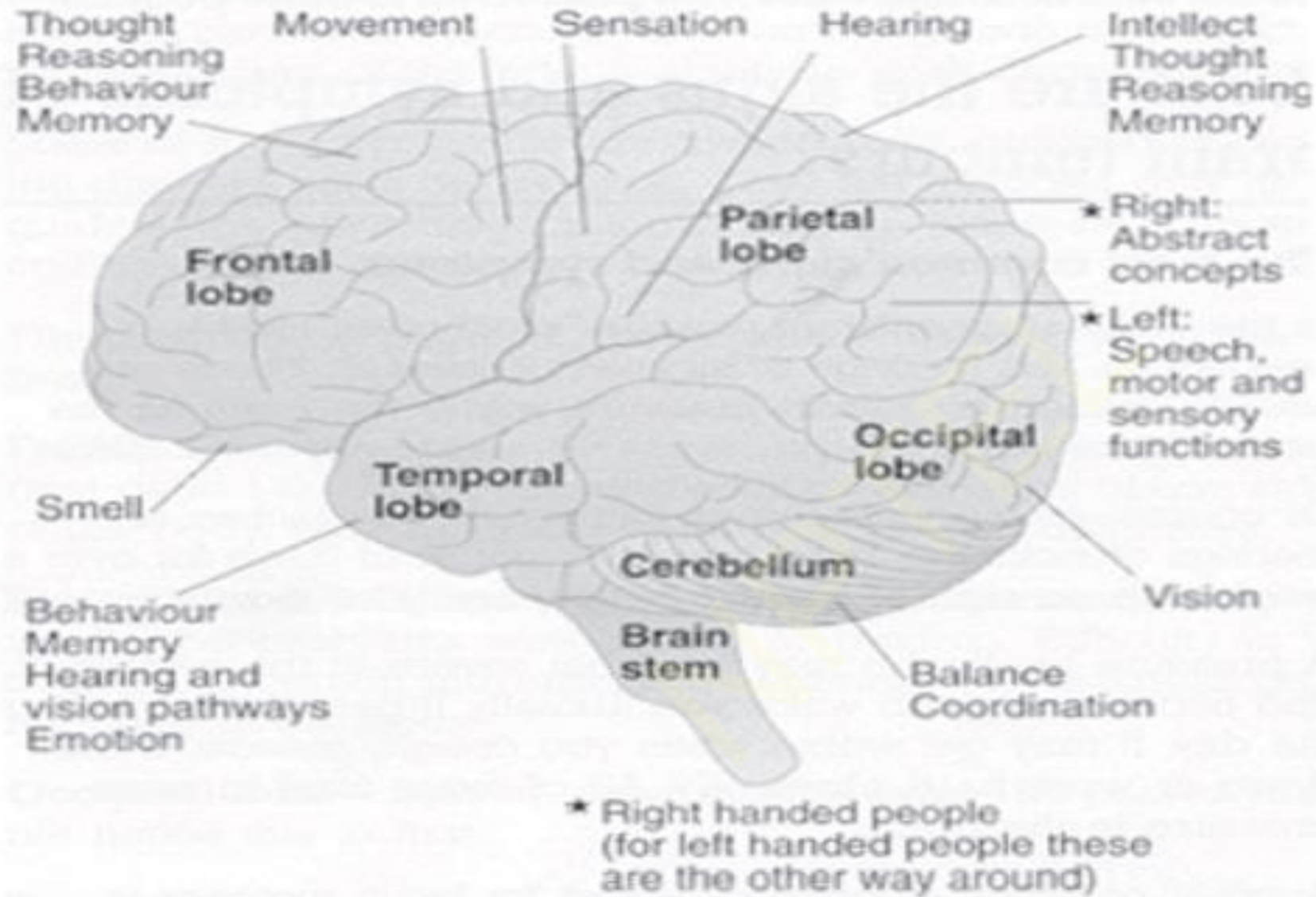
Source: taken from the address of <http://www.educarer.com/brain.html>



Graph 2: The structure of the brain

Source: http://www.brainwaves.com/images/brain-basic_and_limbic.gif





Graph 3: Parts of the brain

Source: <http://www.cancerbacup.org.uk/info/brain/brain-3.htm>

- **BBL** deals with the concept of learning in a neuro-physiological context
- Explains the process of learning as a bio-chemical or electro-chemical change
- Learning, is a process of establishing a connection between brain cells and an intercellular connection; a new connection is established between brain cells with each new learning (Jensen, 2000)
- Structure and processing of brain, intercellular relations, brain hemispheres and lobes (parts), processes at which each part is good at and structural qualities of brain hemispheres are some of the features of BBL
- Brain is a structure that is composed of a number of parallel processors that carry out a lot of processes at the same time

Brain-Based Learning



Brain-Based Learning Theory is based on the structure and function of the human brain. As long as the brain is not prohibited from fulfilling its normal processes, learning will occur.



Table 1: Qualities and strengths of brain hemispheres

Left Brain	Right Brain
It is realistic, logical.	It is intuitional.
It is verbal and numeral dominant.	It has visual and spatial tendencies.
It is planned and structural.	It is spontaneous and natural.
It solves problems through analysis.	It solves problems through synthesis.
It controls feelings.	It lets feelings free.
It ends up with objective judgements.	It ends up with subjective judgements.
It remembers names.	It remembers faces.
It sees details.	It sees the whole.
It likes hierarchically structured interaction.	It likes participative informal interaction.
It tends to think realistically.	It considers feelings while thinking.
It is analytical and logical.	It is creative, reactional and abstract thinker.
It focuses on vertical angle.	It focuses on horizontal angle.
It thinks convergently.	It thinks divergently.
Deduction.	Induction.
It is time-oriented.	It is environment-oriented.
It prefers writing and speaking.	It prefers drawing, using objects and imagining.
It controls right side of the body.	It controls left side of the body.
It deals with one task in a time unit.	It tends to focus on different tasks simultaneously in a time unit.
It tries to find differences.	It tries to find similarities.

Adapted from Jensen, 1998.



- **Brain based learning(BBL)** is a new science that shapes learning process.”Eric Jensen, 2000
- **Brain based learning** is an approach that is based on the structure and function of human brain
Caine and Caine (2002);
- **Making sense in brain** is much more important than receiving knowledge
- Brain creates meanings in line with patterns, connections and feelings
- Jensen (1998),- “**How can brain learn best?**”
- **BBL** focuses on- how brain perceives, operates, interprets, establishes connection, stores (such as establishing connection, coding, structuring matrixes) and remembers messages in learning process



❑ Factors that ensure learning to be meaningful and permanent

1. **Relaxed alertness**: When a person is in an environment where he feels relaxed and calm, learning will be influenced by this environment positively
2. **Orchestrated immersion**: When a certain level of integrity and inter correlativity is ensured, learners will use memory systems in order to explore content and reach learning goals
3. **Active processing**: A learning brain is an active brain; learning activity takes place by means of new connections (dendrites and synapses) established as a result of data transfer between neurons

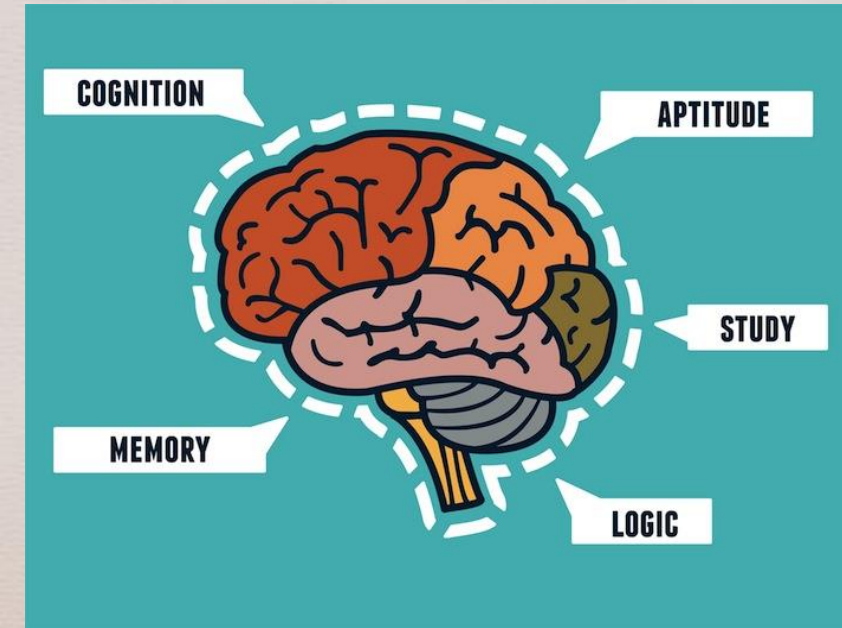


❑ Principles of BBL: How to realize effective learning?

Caine and Caine (1998; 2000)-

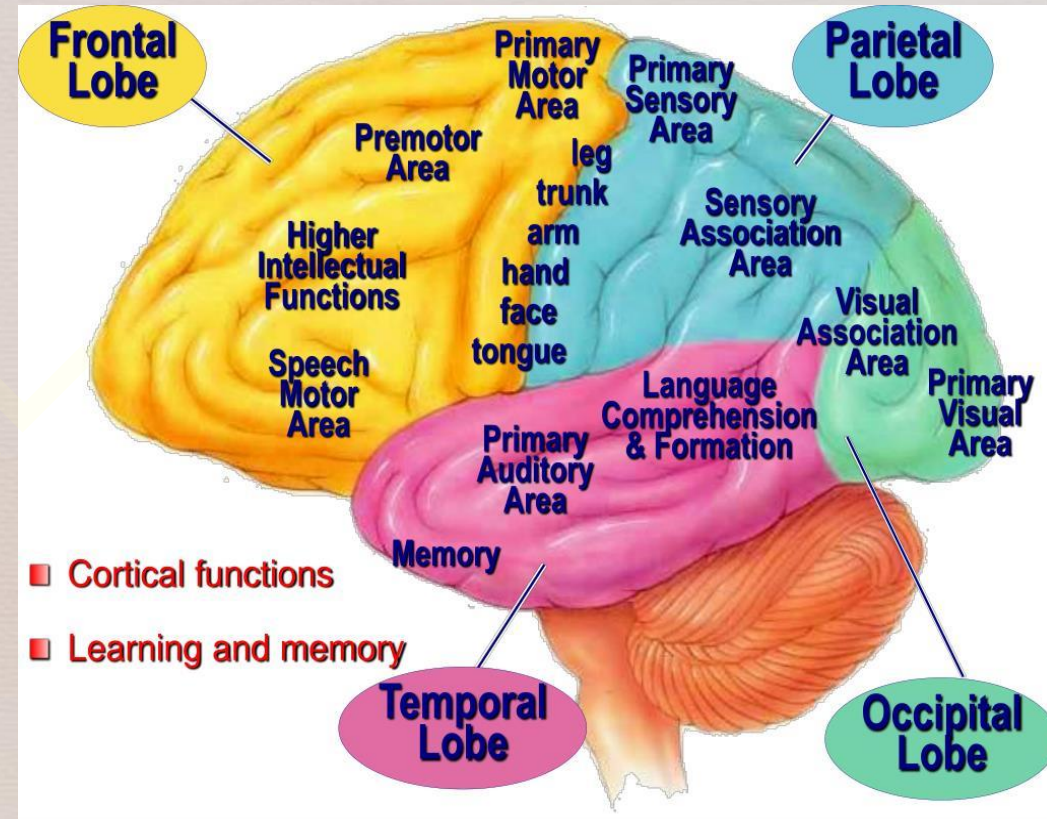
➤ The brain is a parallel processor; learning is also affected by psycho-social qualities of the individual & chemical structure of the organism; these interactions reflect on the process of learning

1. Learning engages the **entire physiology**
2. The search for meaning is **innate**
3. The search for meaning occurs through **patterning**
4. **Emotions are critical** to patterning
5. The brain **processes wholes** and **parts simultaneously**
6. Learning involves both **focused attention** and **peripheral perception**; also affected by **environmental factors**,



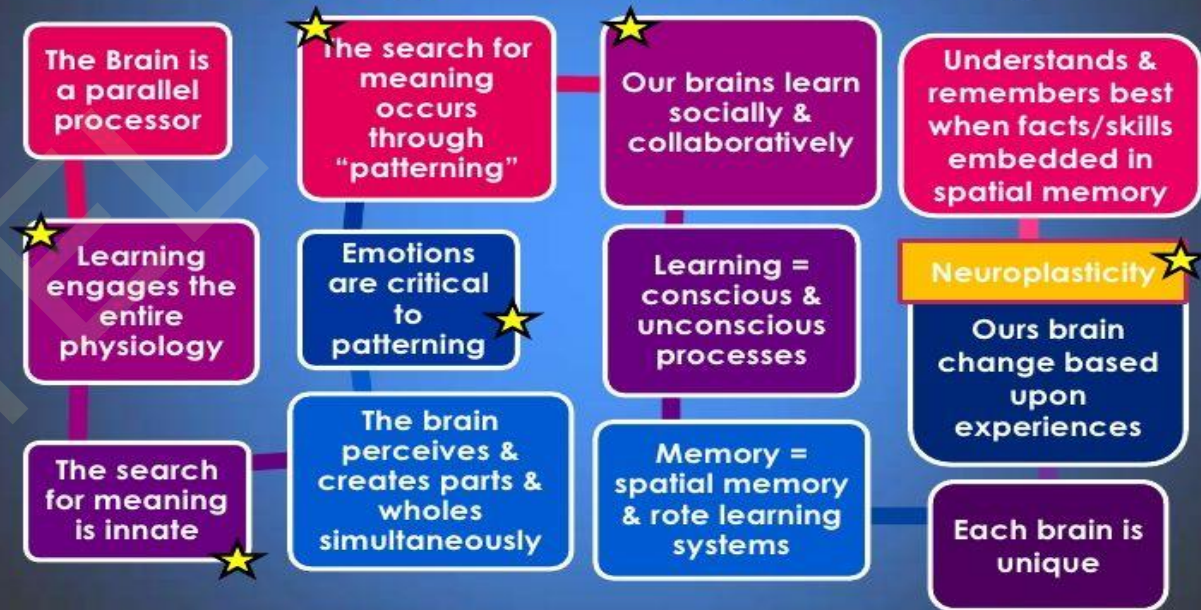


7. Concepts such as awareness, conscious, automatization (unconscious behaviour), feeling, sense & attention are all known to originate from brain and can be called the factors of learning process
 8. Learning involves both conscious and unconscious processes
 9. We have two types of memory: STM ,LTM
 10. We understand and remember best when facts and skills are embedded in natural, spatial memory
 11. Learning is enhanced by challenge and inhibited by threat
 12. Each brain is unique
- ☐ Factors that affect BBL-



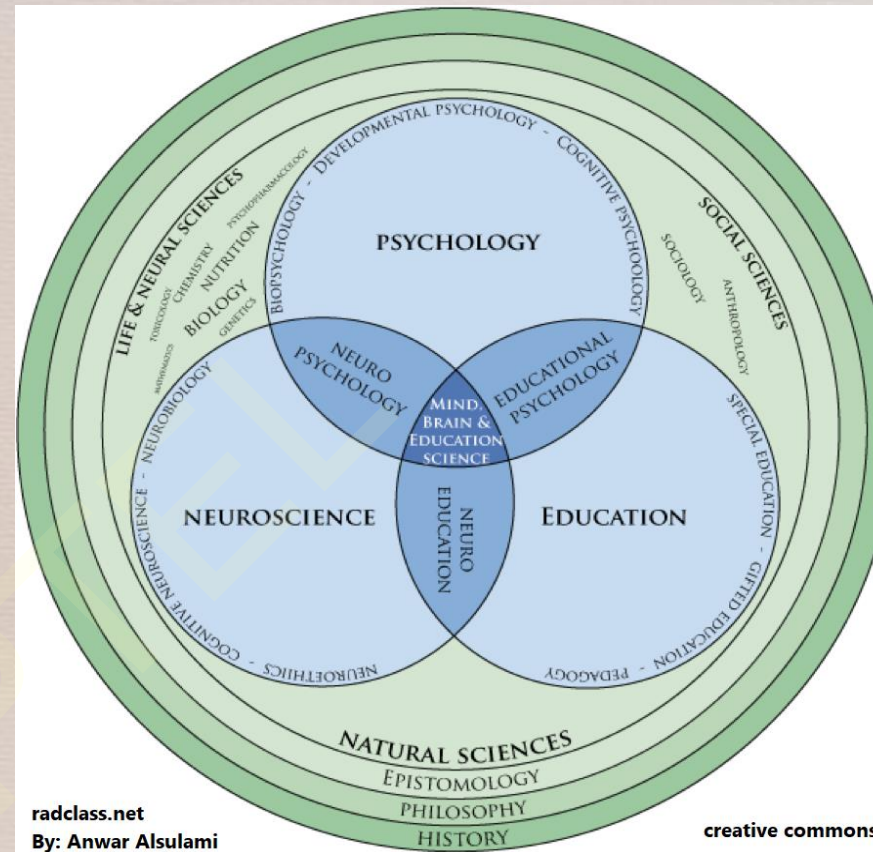
- The organic / chemical structure of the brain
- Emotions
- Music
- Sleep
- Physical vitality
- Exercise
- Nutrition
- Genetic factors
- Lifestyle
- Stress / anxiety
- Motivation
- Social / physical / educational environment
- Health

12 Principles of Brain-Based Learning & Neuroplasticity



- 1 Memory:
- 2 Stress, fear and anxiety
- 3 Attention
- 4 Rewards and punishment
- 5 Social fluency
- 6 Individual development
- 7 Scientific research
- 8 Information literacy
- 9 Artistic expression

□ “Brain based learning is a new science that shapes learning process.” (Eric Jensen, 2000)





Summary

- ✓ Brain needs sufficient nutrition, water and rest to realize best L.
- ✓ Feelings can weaken or strengthen learning
- ✓ New learning is built upon previous learning
- ✓ Brain stores data in more than one place
- ✓ Brain processes the gathered data beyond the awareness of learners
- ✓ Reflection is an important part of learning and processing information
- ✓ Brain has got two memory systems: STM , LTM
- ✓ Brain can acquire information both through a holistic view and in a detailed manner





Lecture:10

“ ”

Major Theories of Learning

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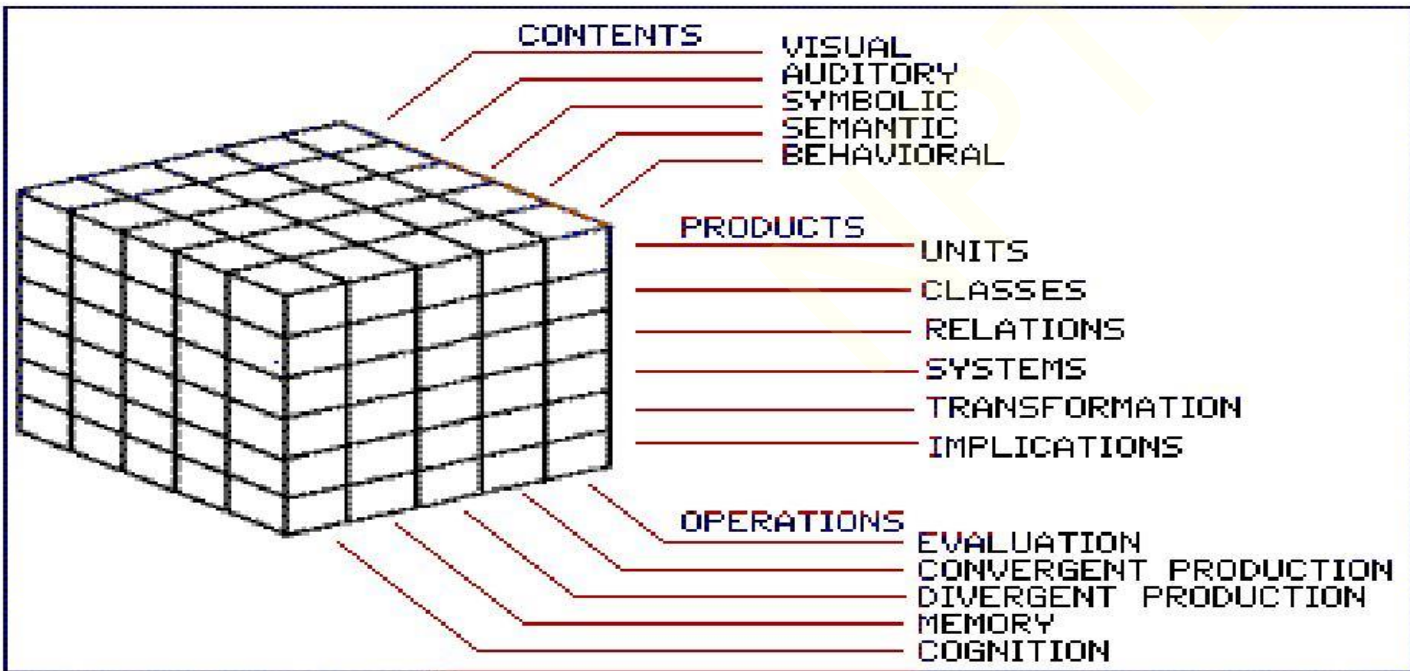
Prof. Atasi Mohanty
Indian Institute of Technology
Kharagpur

NPTEL



According to **Guilford's Structure of Intellect (SI) theory (1955)**, an individual's performance on intelligence tests can be traced back to the underlying mental abilities or factors of intelligence. SI theory comprises up to 180 different intellectual abilities organized along three dimensions: operations, content, and products.

Guilford's Structure-of-Intellect Model



Guilford's Theory of Intelligence

Operations of Guilford's Structure of Intellect Theory

- Evaluation**
Evaluates information to determine its value or quality.
- Convergent Production**
Converges information to a single solution or answer.
- Divergent Production**
Diverges information to multiple solutions or answers.
- Memory**
Stores information for later use.
- Cognition**
Processes information to understand and learn.

Contents of Guilford's Structure of Intellect Theory

- Visual**
Information received through sight.
- Auditory**
Information received through hearing.
- Symbolic**
Information received through symbols.
- Semantic**
Information received through meaning.
- Behavioral**
Information received through action.

Products of Guilford's Structure of Intellect Theory

- Units**
The smallest units of information.
- Classes**
Groups of units that share common characteristics.
- Relations**
Connections between units or classes.
- Systems**
Organized groups of units or classes.
- Transformations**
Changes in the form or structure of information.
- Implications**
Consequences or inferences from information.

Guilford's Structure of Intellect Model

Guilford's Structure of Intellect Model can be visually represented on a three-dimensional cube.

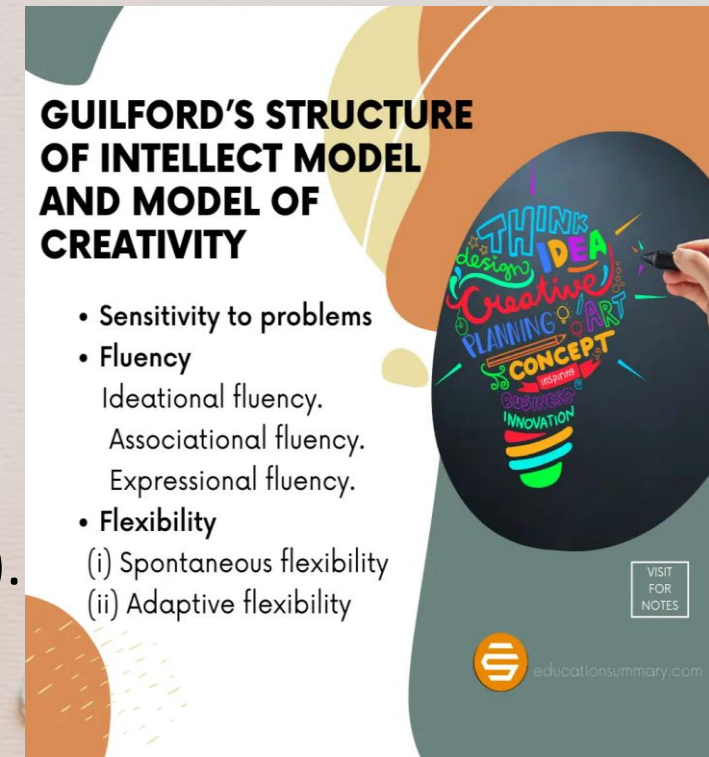
Each cube is composed of 180 smaller cubes, representing the 180 different intellectual abilities.

Each cube is composed of 180 smaller cubes, representing the 180 different intellectual abilities.

Each cube is composed of 180 smaller cubes, representing the 180 different intellectual abilities.



- **Structure of Intellect (J.P. Guilford)**
- In **Guilford's Structure of Intellect (SI) theory**, intelligence is viewed as comprising operations, contents, and products.
- There are **6 kinds of operations** (cognition, memory recording, memory retention, divergent production, convergent production, evaluation),
- **6 kinds of products** (units, classes, relations, systems, transformations, and implications), and
- **5 kinds of contents** (visual, auditory, symbolic, semantic, behavioral).
- Since each of these dimensions is independent, there are theoretically **180 different components of intelligence**.
- Guilford researched and developed a wide variety of psychometric tests to measure the specific abilities predicted by SI theory.
- A major impetus for Guilford's theory was his interest in creativity (**Guilford, 1950**).
- The **divergent production operation** identifies a number of different types of creative abilities.

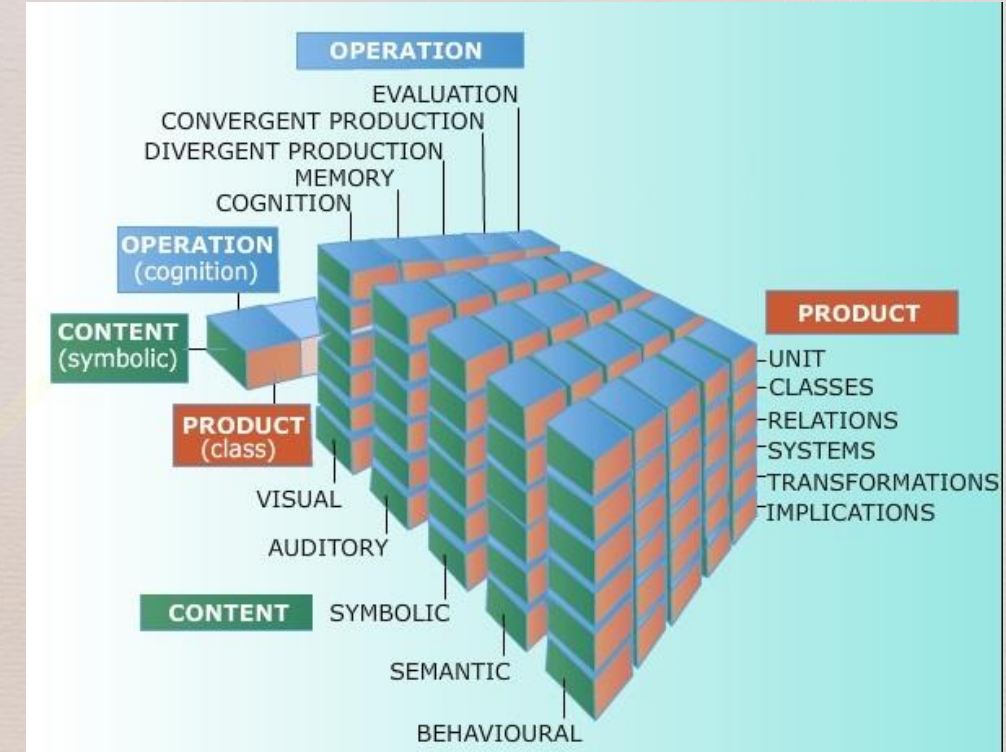




- SI theory is intended to be a general theory of human intelligence; its major **application** has been in education, personnel selection and placement.

- **Principles**

- **Reasoning and problem-solving skills** (convergent and divergent operations) can be subdivided into 30 distinct abilities (6 products x 5 contents).
- **Memory operations** can be subdivided into 30 different skills (6 products x 5 contents).
- **Decision-making skills** (evaluation operations) can be subdivided into 30 distinct abilities (6 products x 5 contents).
- **Language-related skills** (cognitive operations) can be subdivided into 30 distinct abilities (6 products x 5 contents).





- **Gagne's Conditions Of Learning Theory**
- Gagne's **Conditions of Learning**, also known as **Nine Events of Instruction**, is a set of **instructional design principles** developed by psychologist Robert Gagne.
- These conditions outline **a sequence of events that enhance the learning process and promote effective instruction.**
- Gagne's theory of instructional learning offers a more rounded **theory of learning** and **instruction** that which offers far more directive advice on **how the teacher should manage individual lessons.**
- The nine events provide **a framework for designing and delivering instruction in a structured and systematic manner.** Such as :



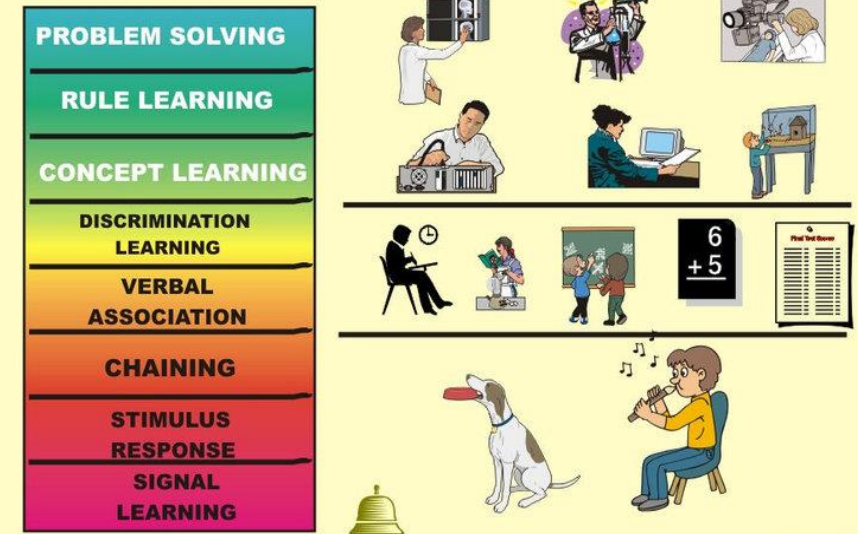
❑ Theory of Learning & Instruction: Robert Gagne



GAGNE'S NINE EVENTS of Learning & Instruction:

- 1. Gain attention of the students
- 2. Inform students of the objectives -
 - ✓ Describe required performance
 - ✓ Describe criteria for standard performance
 - ✓ Learner establishes criteria for standard performance
- 3. Stimulate recall of prior learning
- 4. Present the content
- 5. Provide learning guidance
- 6. Elicit performance (practice)
- 7. Provide feedback
- 8. Assess performance
- 9. Enhance retention and transfer to the job

Gagne's Hierarchy of Learning



Sprauls



- **Gain Attention:** This event aims to **capture the learners' attention** and **engage them** in the learning process.
- It can be achieved through the use of stimulating and relevant stimuli or by posing questions or problems.
- **Inform Learners of the Objective:** **Learners need** to be aware of the **specific learning objectives** or goals they are expected to achieve.
- Clear communication of these objectives helps **to focus their attention** and **motivate them to learn**.
- **Stimulate Recall of Prior Knowledge:** **Activating learners' prior knowledge** helps them **connect new information** to existing mental frameworks.
- By **reviewing relevant concepts** or experiences, learners can build upon what they already know.

9 Events of Instruction

- #1 • Gain attention (the hook)
- #2 • Orient the learner (prep and objectives)
- #3 • Stimulate recall of prior knowledge
- #4 • Present content material
- #5 • Provide learner guidance
- #6 • Elicit performance "practice"
- #7 • Provide informative feedback
- #8 • Assess if lesson objectives have been learned
- #9 • Enhance retention and transfer



- **Present the Content:** The instructional content is presented to the learners in a structured and organized manner.
- It should be logically sequenced, chunked into manageable units, and delivered using appropriate instructional strategies such as lectures, visuals, or multimedia.
- **Provide Guidance:** Learners need guidance and support to understand and acquire new knowledge or skills.
- This event involves providing clear explanations, examples, demonstrations, and instructions to assist learners in grasping the content.
- **Elicit Performance:** Learners are given opportunities to practice what they have learned.
- This active participation helps reinforce the newly acquired knowledge or skills and allows for feedback and correction if needed.

Intellectual Skills	• Problem solving, discriminations, concepts, principles
Cognitive Strategy	• Meta-cognition - strategies for problem solving and thinking
Verbal Information	• Facts of knowledge
Attitude	• Actions that a person chooses to complete
Motor Skills	• Behavioral physical skills



- **Provide Feedback:** Learners receive feedback on their performance, indicating whether they have achieved the desired learning outcomes.
- Feedback helps them **assess their progress**, identify areas for improvement, and **reinforce correct** understanding or **behavior**.
- **Assess Performance:** This event involves assessing **learners' performance** to determine the extent to which they have achieved the learning objectives.
- Various **assessment methods** such as quizzes, tests, or practical exercises, can be used to **evaluate their progress**.
- **Enhance Retention and Transfer:** The final event focuses on promoting **long-term retention** and **transfer of the learned material** to real-world contexts.
- **Strategies** such as providing opportunities for review, application in different situations, and promoting transfer of knowledge to other domains are **employed to solidify learning**.

Learning Domains : 5 types of learning outcome

Outcome or Ability	Meaning	Performance example
*Intellectual skill	Ability to interact with surroundings using concept symbols	Identify the diagonal of a square. Explaining why ice freezes at 0°C. Predicting the rate of growth of a plant based on conditions of water, soil, light, etc.
Cognitive strategy	Ability to control the individual's behaviour to learn, remember & think.	To draw a chart for organizing data. To reason backwards to solve problem. Breaking up a problem into various parts.

What Is Mastery Learning?

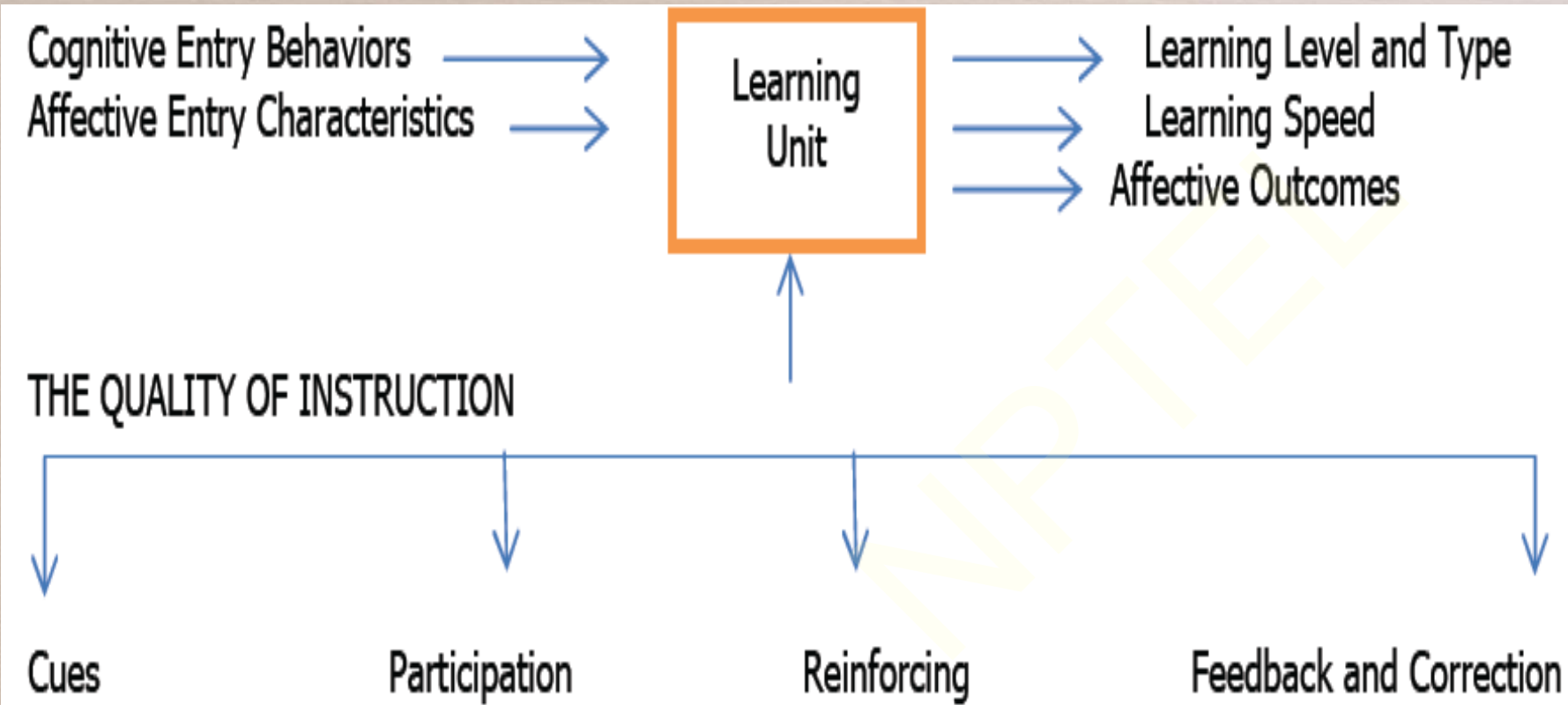


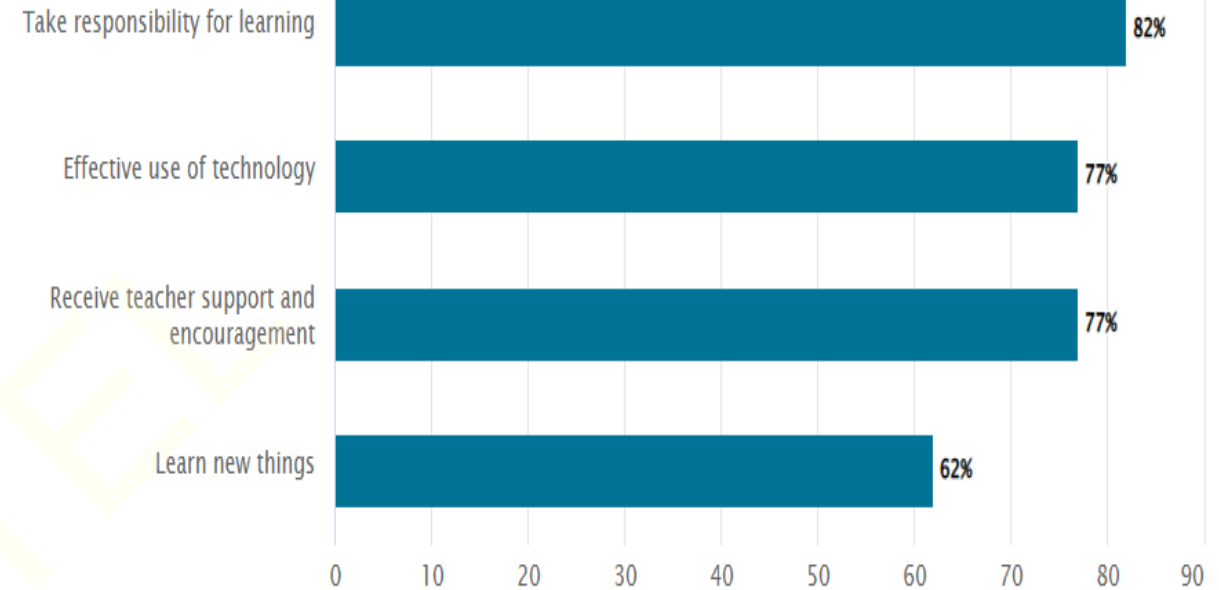
Figure 1: Mastery Learning

How does Mastery Learning work?



Research.com

Effects of Mastery Learning on Students



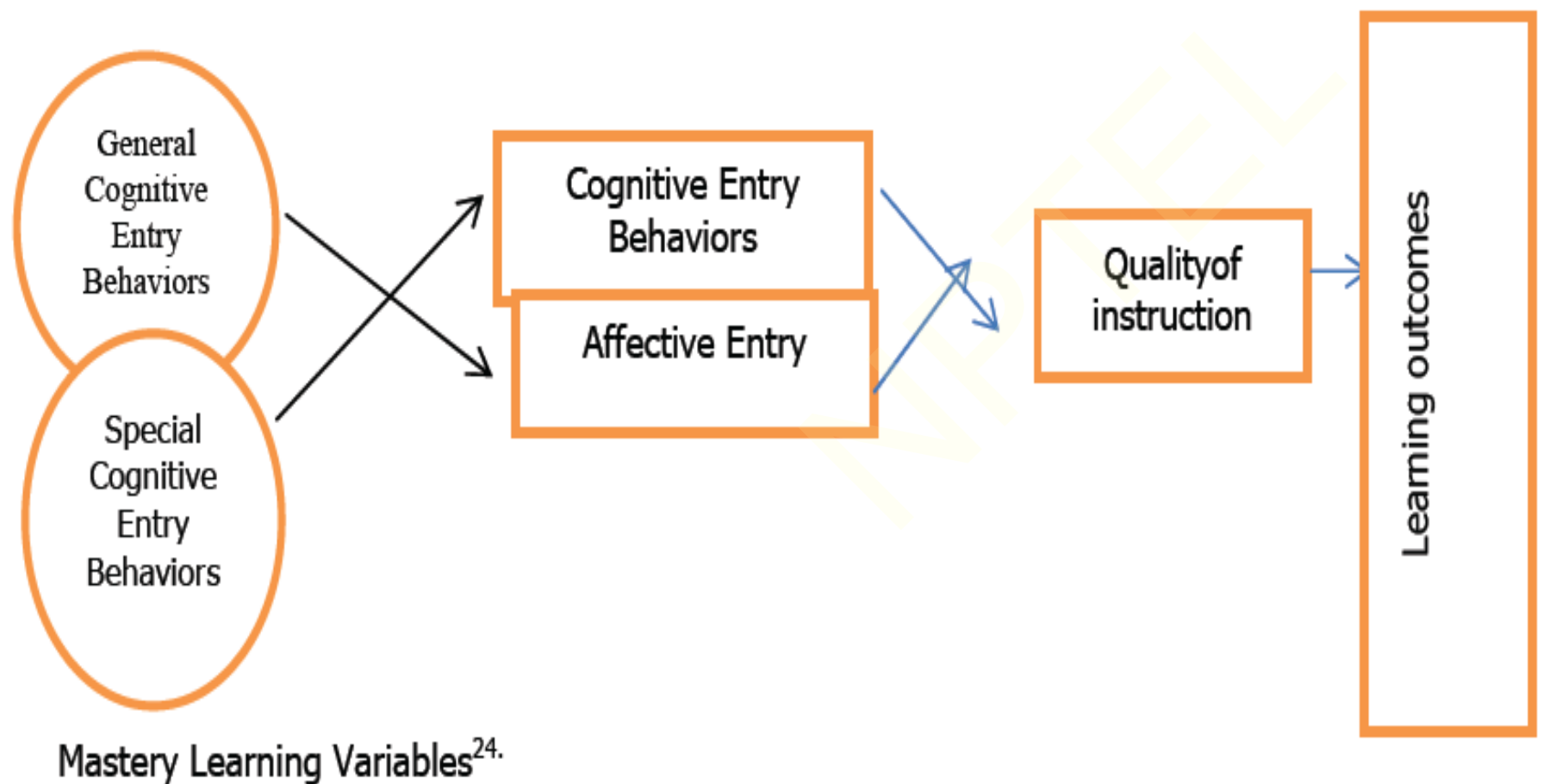
Barnett (2020)

Designed by Research.com



➤ 1. Cognitive Entry Behaviours: language ability, & reading comprehension

➤ 2. Affective Entry Characteristics: interests, attitudes & learning readiness



• Elements of Mastery Learning

➤ Pre-assessment-

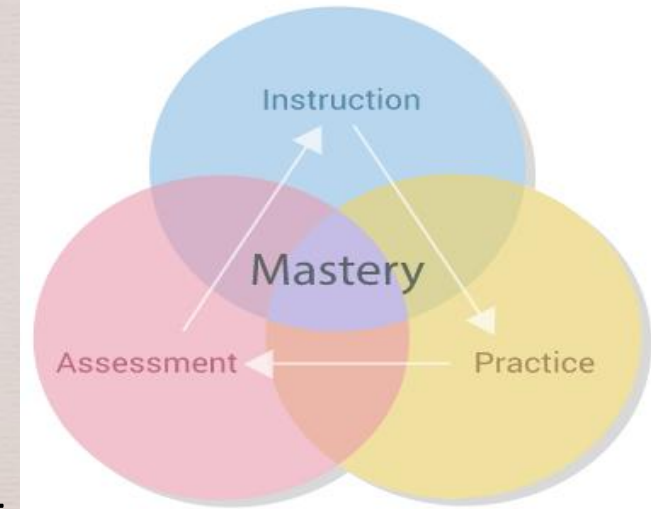
- ✓ Determining students' prior knowledge and skills
- ✓ Monitoring student progress
- ✓ Communicating expectations
- ✓ Focusing students' attention on learning targets
- ✓ Checking for misconceptions
- ✓ Identifying students' interests, talents, and preferred ways of learning

➤ Group-based initial instruction

- ✓ Instruction must be multifaceted, context-adapted, and student-oriented (Guskey, 2010)

➤ Regular formative assessments

- ✓ The use of regular formative assessments.
- ✓ This could be in the form of quizzes, written assignments, oral presentations, skill demonstrations, or performances, depending on the subject area, the grade level, and the learning outcomes involved





➤ Corrective Instruction

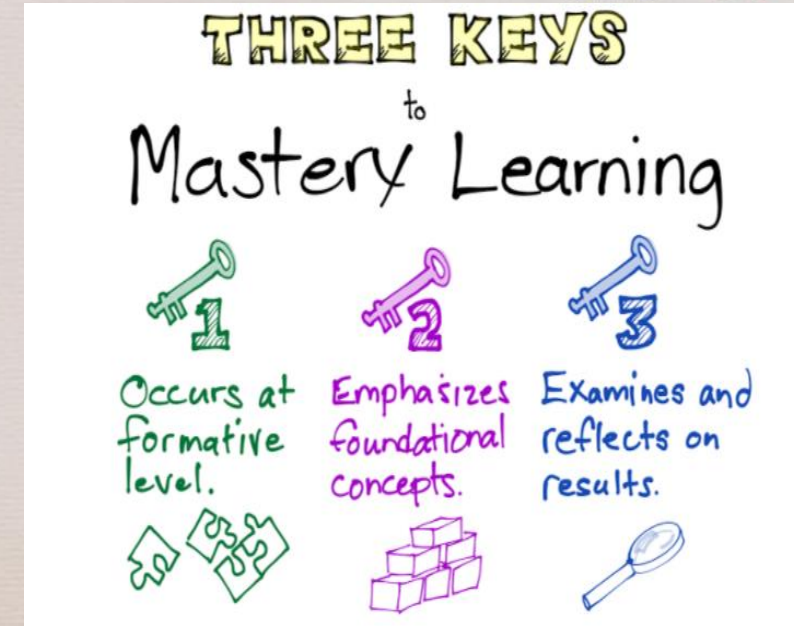
- ✓ Teachers provide “**high-quality corrective instructions**” to mend learning problems that the assessments have identified,
- ✓ **CI** to be versatile enough to accommodate different student **learning styles, modalities, and intelligence levels**

➤ Parallel formative assessments

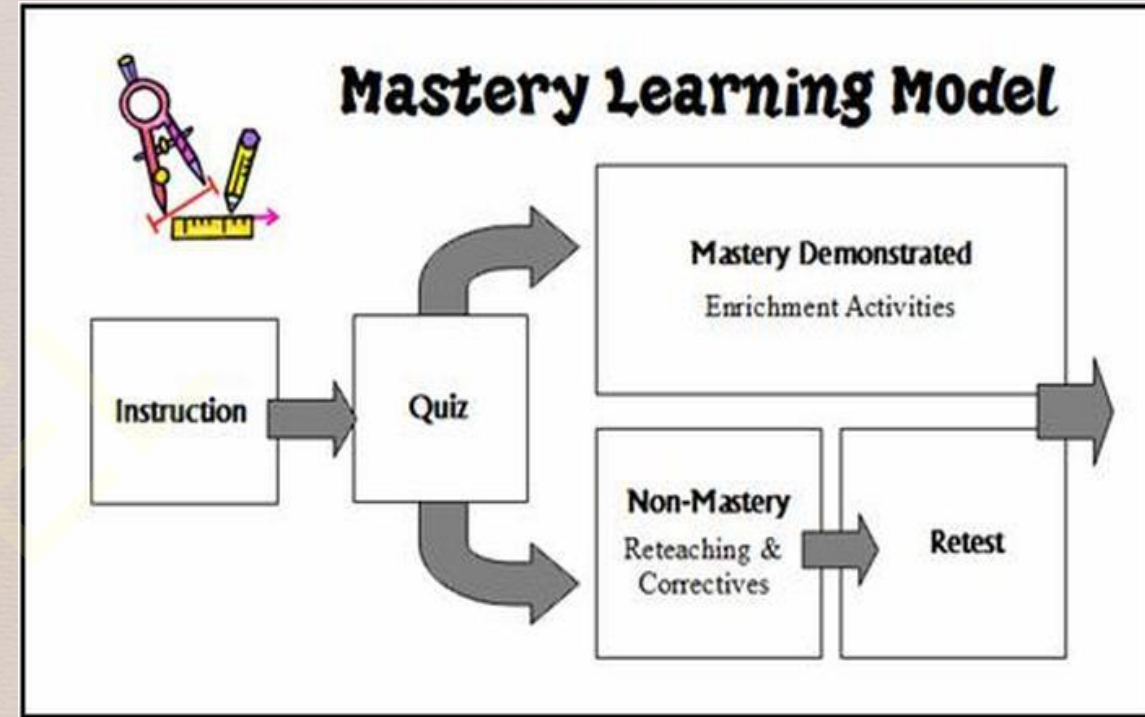
- ✓ A second, **parallel formative assessment** to determine the **effectiveness** of the **corrective instruction**.

✓ Enrichment activities

- ✓ Aiming to provide challenging yet rewarding learning experiences, these activities enable students **to explore a greater depth of related topics** that pick their interests.
- ✓ Enrichment activities could be in the form of **academic games** and exercises, various **multimedia projects**, and **peer discussions**.



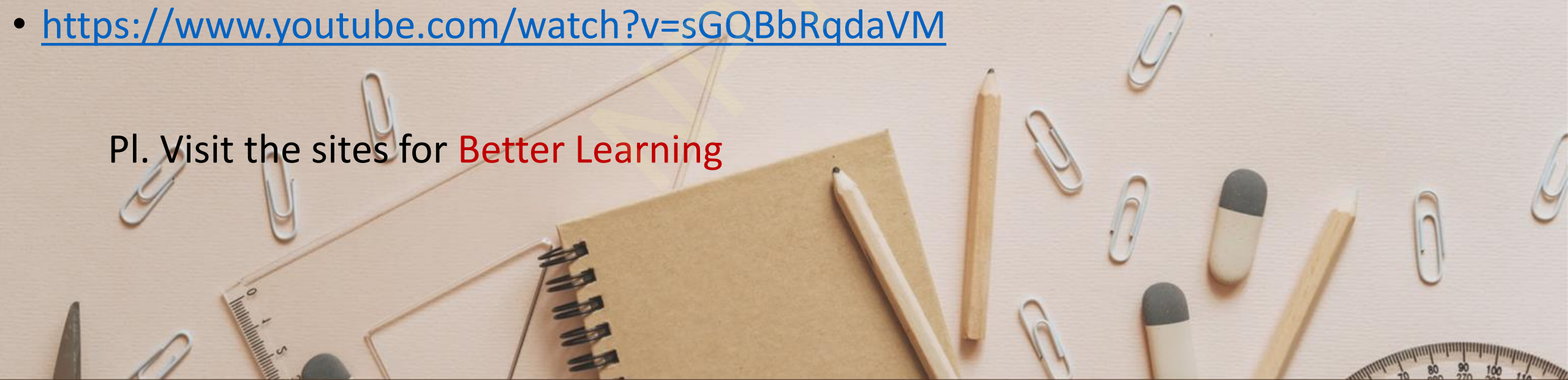
- **Concept of Personalization and Academic Self-Concept**
- **QUALITY OF INSTRUCTION**
 - Cues, Participation, Reinforcing
 - Feedback and Correction
- **LEARNING OUTCOMES**
 - Learning Level and Learning Speed
 - Affective Outcomes
- **POSITIVE SIDES OF MASTERY LEARNING:**
 - 1. student learns in his own capacity and speed,
 - 2. increases the achievements on affective and cognitive level





- <https://www.youtube.com/watch?v=efjDq-kO7lE>
- <https://www.youtube.com/watch?v=GWa48XRnLh0>
- <https://www.youtube.com/watch?v=tBiz04uGqA4>
- <https://www.youtube.com/watch?v=ej1qP8o2Sp4>
- <https://www.youtube.com/watch?v=N1ESb1RGZ7g>
- <https://www.youtube.com/watch?v=sGQBbRqdaVM>

Pl. Visit the sites for **Better Learning**





Constructivism

Jean Piaget (1896-1983),
Lev Vigotsky (1896-1934)



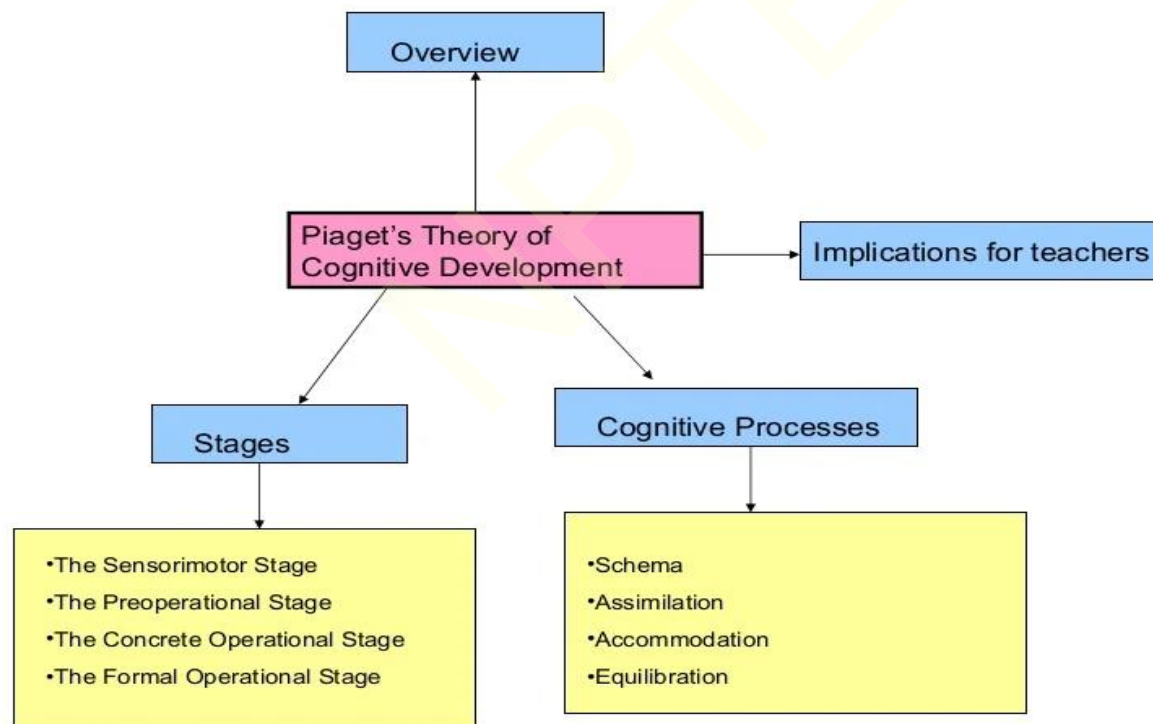
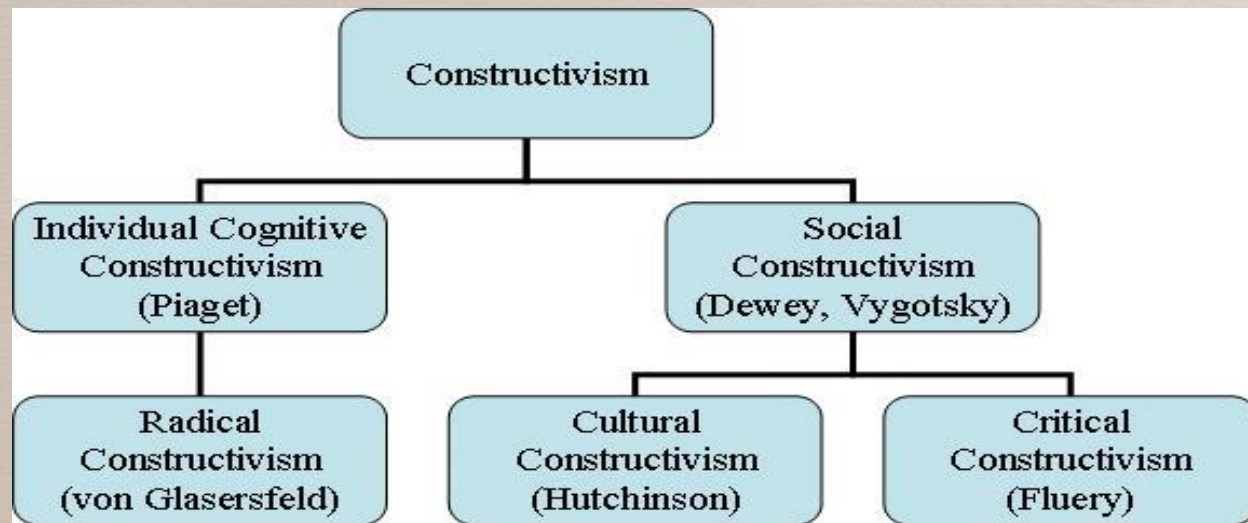
- **Constructivism** is ‘an approach to learning that holds that people actively construct or make their own knowledge and that reality is determined by the experiences of the learner’ (Elliott et al., 2000).
- **Constructivism** is crucial to understand as an educator because it influences the way all of our students learn.
- Teachers and instructors who understand the constructivist learning theory understand that their students bring their own unique experiences to the classroom every day.
- Their background and previous knowledge impacts how they are able to learn.





- **Types of constructivism-**

- **1. Cognitive-** Cognitive constructivism focuses on the idea that **learning should be related to the learner's stage of cognitive development.**
- It comes from the work of **Jean Piaget** and his research on cognitive development in children.
- **2. Social-** Social constructivism focuses on the collaborative nature of learning. **Knowledge develops from how people interact with each other,** their culture, and society at large.
- **Lev Vygotsky**, and is closely connected to cognitive constructivism with the added element of **societal and peer influence.**
- **3. Radical-** Radical constructivism is very different from cognitive and social constructivism. It focuses on the idea that **learners and the knowledge they construct tell us nothing real, only help us function in our environment.**
- This theory was developed by Ernst von **Glaserfeld** in 1974.



- **Constructivism: A Learning Theory**
- Constructivism is a learning theory that says **we create our own knowledge from our experiences and interactions**,
- It challenges the idea that knowledge is fixed and objective, and that teachers can simply transfer it to students,
- Constructivists see **learning as a dynamic and social process**, where **learners build on their prior knowledge** and **collaborate with others**,
- Constructivism **encourages learners to be active, creative, reflective, and critical thinkers** who can construct their own understanding of the world,



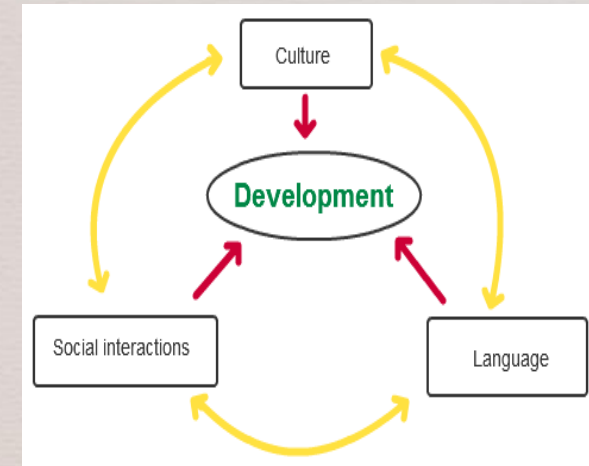
WHAT IS CONSTRUCTIVISM?

Construct = to build

Constructivism is the theory which states that knowledge cannot exist outside the mind of the pupils, but it is **built** in the mind based on real

❑ Basic Concepts of Constructivism

- Cognition (learning) is the result of mental construction
- People construct their own understanding and knowledge of the world, through experiencing things & reflecting on those experiences
- Human learning is constructed; learners build new knowledge upon the foundation of previous learning
- Constructivist learning is transferable
- Constructivism gives students ownership of what they learn
- Students in constructivist classrooms learn to question things
- Promotes social & communication skills through collaboration and exchange of ideas





Cognitive constructivism is based on the work of **Jean Piaget** & **Social constructivism** is based on the work of **Lev Vygotsky**.

❑ **Principles of Constructivism:** (Tam ,2000)

- **Knowledge** will **be shared** between teachers and students
- Teachers and students will **share authority**
- The **teacher's role** is one of a **facilitator or guide**
- Learning **groups will consist of** small numbers of **heterogeneous** students
- To **embed learning** in social experience, (**collaboration**) & **realistic context**
- To **encourage awareness** of the **knowledge construction process** (**reflection, metacognition**)





- Learning is an individual and social process.
- Learning is an organizational process - new knowledge is assimilated with old knowledge to construct new meanings and understandings.
- Learning is based on personal experiences - different people understand 'truth' differently.
- The concept of reality is based on individual interpretation.
- Learning is socially situated and enhanced through meaningful context - situated learning focuses on creating meaning from the real activities of daily life.
- Language plays an essential role in learning - the sharing of knowledge happens through communication.
- Motivation is key to learning - individuals must want to actively engage and reflect on their prior knowledge in learning.

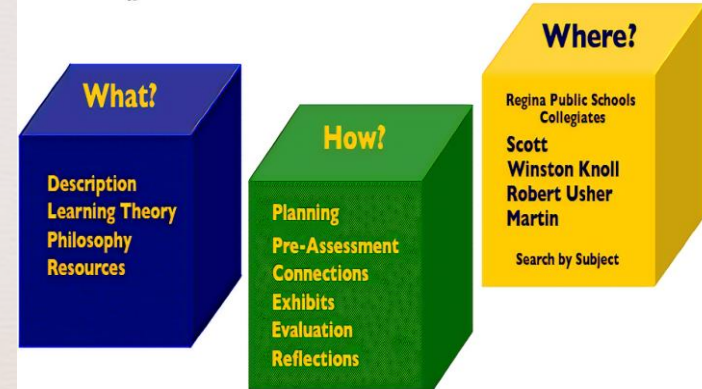
❑ Jean Piaget (1896-1983): Learning and Cognitive Development

- ✓ Knowledge is actively constructed
- ✓ In the process of constructing knowledge, children adhere to their experiences
- ✓ Knowledge is constructed in schemas
- ✓ After new learning, these schemas are organized & developed better and new connections are established
- ✓ Learning & cognitive development occurs with processes of assimilation and accommodation
- ✓ Knowledge is subjective because people construct it
- ✓ Cognitive development occurs step by step



Constructivism

Knowledge Building in the Secondary Classroom

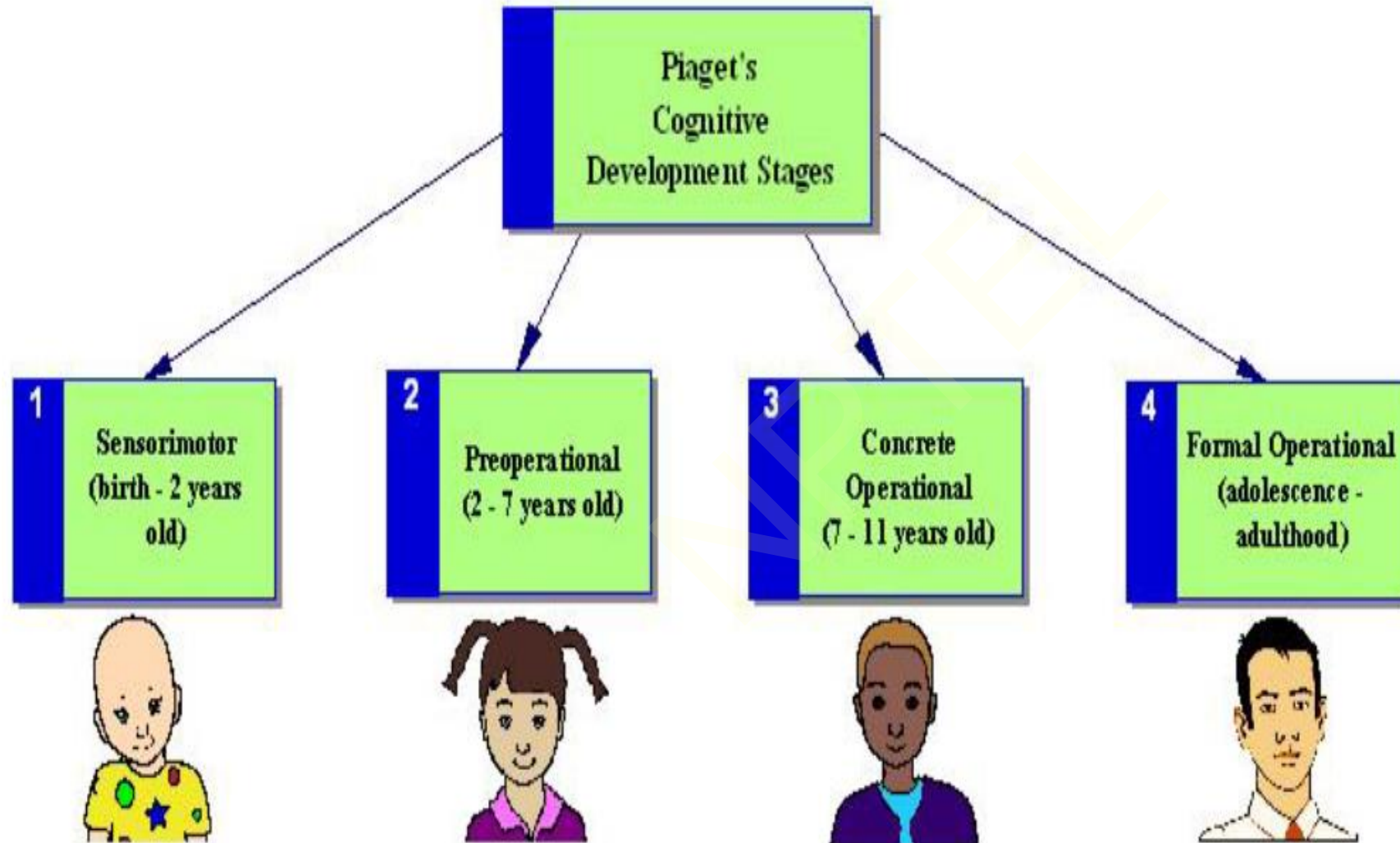


Jean Piaget

- children's schemes change over time
- cognition develops in stages, not gradually
- each stage builds on accomplishments of prior stage
 - Sensory motor
 - Preoperational
 - Concrete operational
 - Formal operational

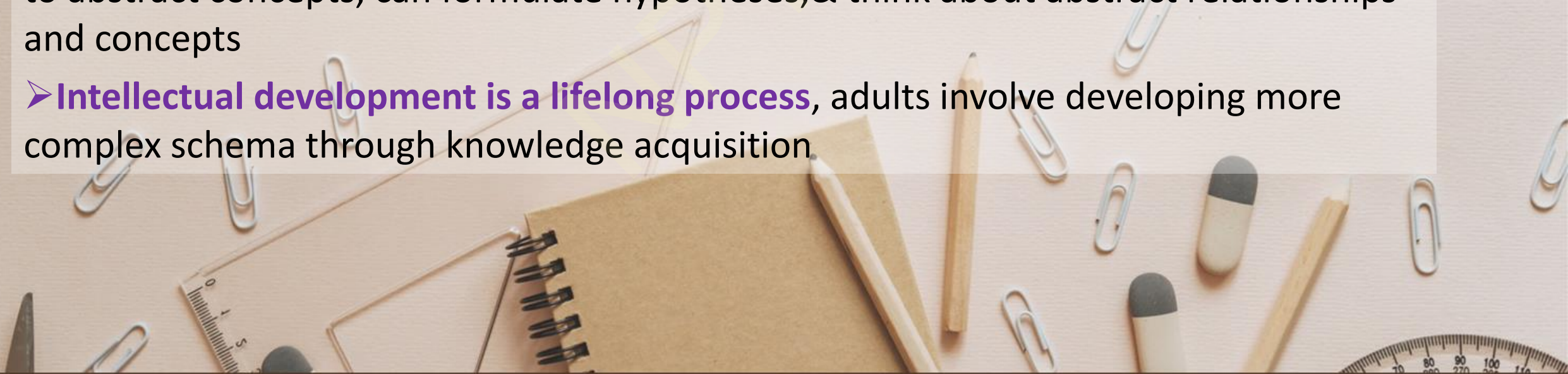


Stages of Cognitive Development

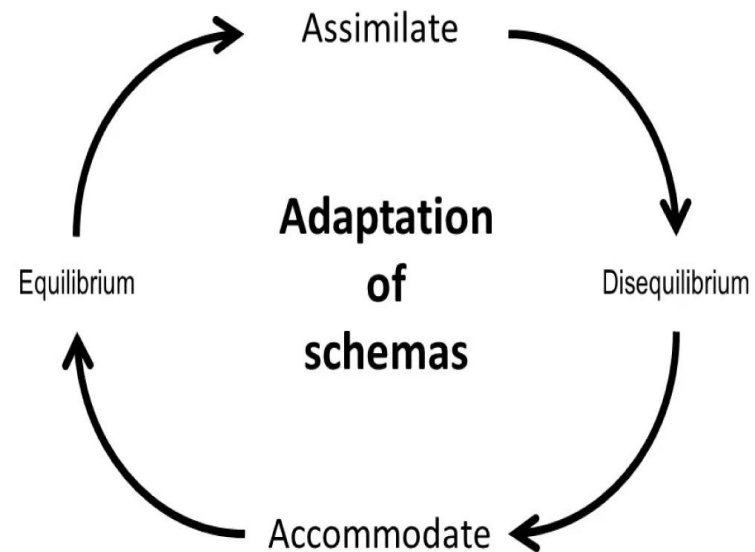
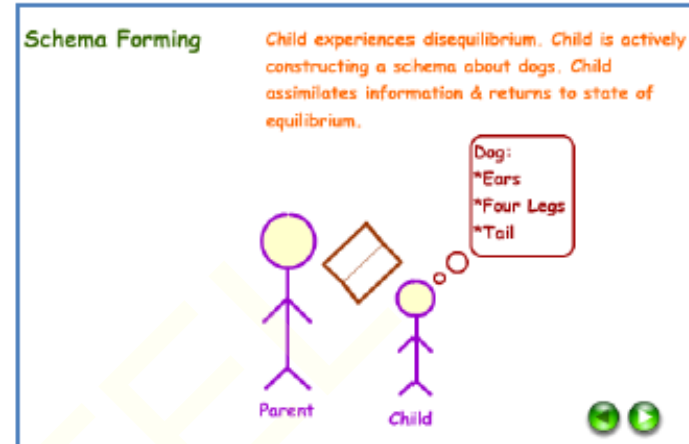
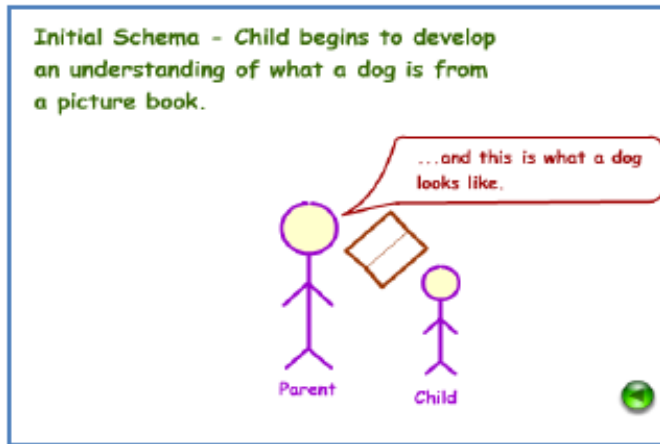




- **Preoperational stage:** children begin to use language; memory and imagination; Intelligence is egocentric and intuitive, not logical
- **Concrete operational stage:** use of logical and systematic manipulation of symbols; Thinking becomes less egocentric with increased awareness of external events; involves concrete references
- **Formal operational stage:** use symbols related to abstract concepts, can formulate hypotheses, & think about abstract relationships and concepts
- **Intellectual development is a lifelong process,** adults involve developing more complex schema through knowledge acquisition.



Schema Formation: Assimilation & Accommodation



What is cognitive constructivism?

- Theory developed by Jean Piaget
- Learning is an active process of **assimilation** and **accommodation**



Active Experience



Making errors



Looking for solutions

Piaget's Stages of Cognitive Development

Formal Operational Stage (12 – 18 years)

- Children form inductive and deductive reasoning. They can use abstract thought and general principles to develop increasingly complex hypotheses.

Concrete Operational Stage (7 – 12 years)

- Logical thought: children begin to see relationships between mass, time, space, etc.
- Conservation: Children discover that changes in appearance do not necessarily correspond with changes in weight, volume, etc.

Preoperational Stage (2 – 7 years)

- Symbolic Thought: Young children learn to use language to represent their thoughts. They develop imaginative play.
- Egocentrism: Young children struggle to see things from other perspectives

Sensorimotor Stage (0 – 2 years)

- Object Permanence: Babies learn that objects out of sight still exist.
- Goal Directed Action: Babies learn to act intentionally to achieve a goal.
- Deferred Imitation: Babies continue to imitate others after the event.





“**Constructivism** is a theory in education which posits that **individuals or learners** do not acquire knowledge and understanding by passively perceiving it within a direct process of knowledge transmission, rather they **construct new understandings and knowledge through experience and social discourse**, integrating new information with what they already know (prior knowledge)”.

➤ **Educational Implications of Constructive Learning-**

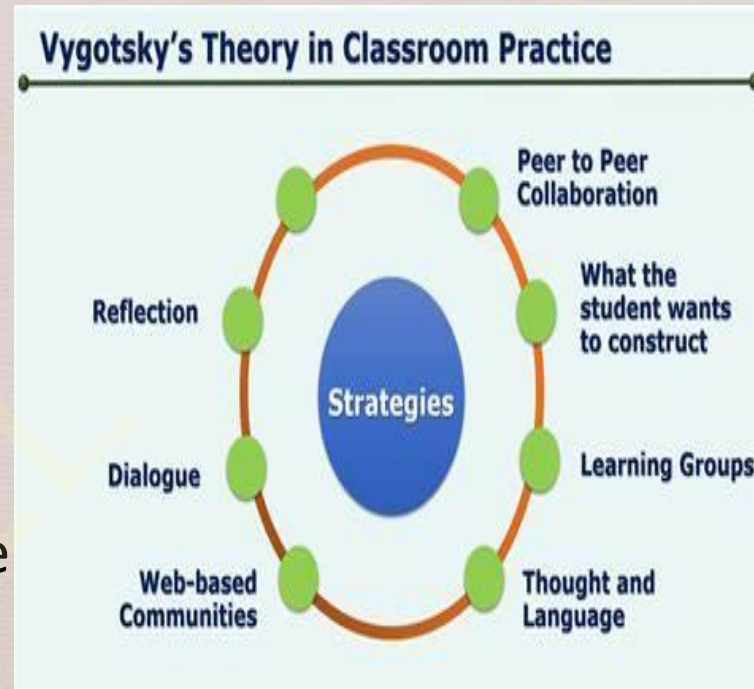
- ✓ **Learners are active participants** in their learning, and learning by doing, or experiential learning is central to constructivist learning in practice (Howe et al., 2000)
- ✓ **Learners are self-regulated** and they construct and monitor their learning, where metacognition plays an important role in meaningful learning.
- ✓ **Social interactions is essential** for meaningful learning.



- Vygotsky's constructivism is a **theory of child development** that emphasizes the **role of social interaction** and **collaboration in learning**.
- Vygotsky argued that **cognitive functions** are the **products of social negotiations** and that learning occurs in the **zone of proximal development**, which is the gap between what a learner can do independently and what they can do with the help of a more knowledgeable other.
- Vygotsky also **rejected the idea** that **learning can be separated from its social context**.
- Social constructivism **shifts the responsibility of knowledge acquisition** from the teacher to the student,
- **Transforms the student** from a passive listener to an **active participant** and a **co-creator of knowledge** among co-learners.

❑ Lev Vygotsky (1896-1934):

- Social interaction leads to continuous step-by-step changes in children's thought & behaviour that can vary greatly from culture to culture
- Development depends on interaction with people and the tools that the culture provides to help form their own view of the world
- Three ways a cultural tool can be passed from one individual to another-
 - ✓ a) Imitative learning, where one person tries to imitate or copy another
 - ✓ b) Instructed learning which involves remembering the instructions of the teacher & then using these instructions to self-regulate the learning behaviour.





- ✓ c) **Cultural tools** are passed on to others is through **collaborative learning**, which involves a group of peers who strive to understand each other and work together **to learn a specific skill**

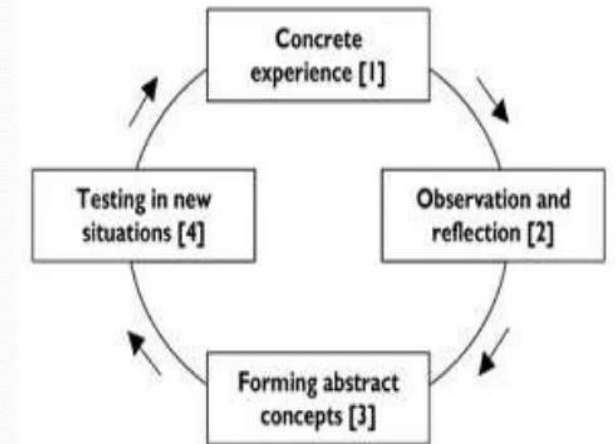
Key points in Vygotsky's theory.

- Social interactions are critical; knowledge is coconstructed between two or more people.
- Self-regulation is developed through internalization (developing an internal representation) of actions and mental operations that occur in social interactions.
- Human development occurs through the cultural transmission of tools (language, symbols).
- Language is the most critical tool. Language develops from social speech, to private speech, to covert (inner) speech.
- The zone of proximal development (ZPD) is the difference between what children can do on their own and what they can do with assistance from others. Interactions with adults and peers in the ZPD promote cognitive development.

(Meece, 2002)

- Vygotsky's position is a form of **dialectical (cognitive) constructivism** because it emphasizes the **interaction between persons and their environments**; **Mediation** is the key mechanism in development and learning
- A key concept is the **Zone of proximal development (ZPD)**, defined as “the distance between the actual developmental level as determined by independent problem solving and the level of potential development as determined through problem solving under **adult guidance or in collaboration with more capable peers**” (Vygotsky, 1978).

Constructive Learning Model



10/14/2015

Parshuram Sharma

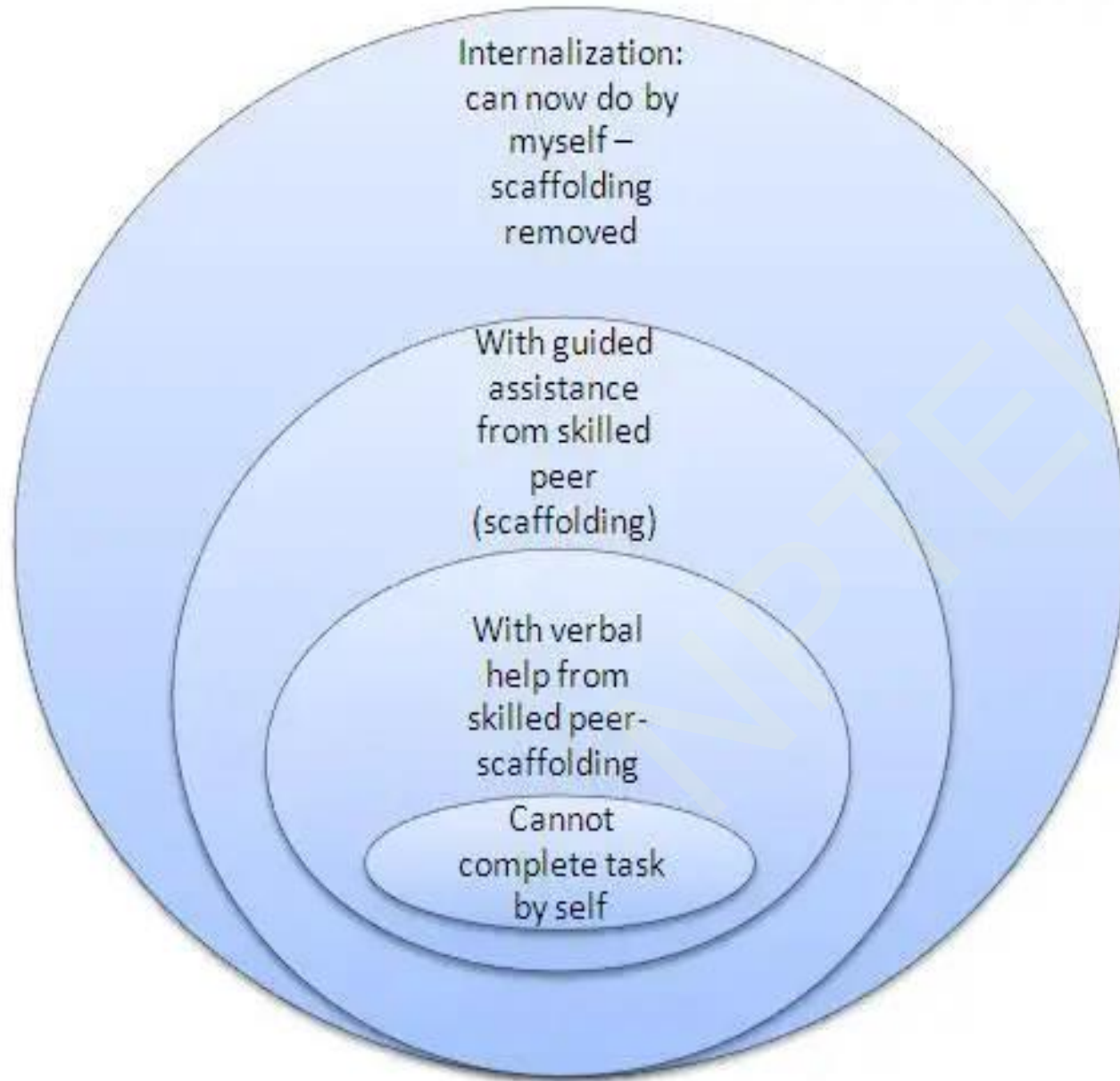
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- **Schooling** is important not because it is where children are **Scaffolded** but, because it allows them to develop **greater awareness of themselves, their language**, & their role in the world order
- Helping students acquire **cognitive mediators** (e.g., **signs, symbols**) through the social environment can be accomplished in many ways
- Concept of **instructional scaffolding**, which refers to the **process of controlling task elements** that are beyond the learners' capabilities so that they can focus on and **master those features** of the task that they can grasp quickly
- **Reciprocal teaching** involves an **interactive dialogue** between a teacher and small group of students.



- **Peer collaboration**, which reflects the notion of collective activity
- **Apprenticeships**- novices work closely with experts in **joint work-related activities**
- Many mental processes are acquired as a result of **social interactions**
- Successful people's help let students reach difficult targets
- The difficulties encountered in achieving the works contribute to the **highest level of cognitive development**
- **Games**, for children, aren't waste of time but rather they include activities that **get them into adults' world successfully**, through **future planning**







- <https://www.khanacademy.org/test-prep/mcat/processing-the-environment/cognition/v/piagets-stages-of-cognitive-development>
- <https://www.youtube.com/watch?v=2qVHU8CTIM8>
- https://www.youtube.com/watch?v=hcXkp0c_q4c
- <https://educationaltechnology.net/constructivist-learning-theory/>
- <https://edpsych.pressbooks.sunycreate.cloud/chapter/social-constructivism-vygotskys-theory/>

